UNIT - I : STRUCTURE OF METALS AND CONSTITUTION OF ALLOYS

1. Introduction

* Metals are crystalline solids: atoms are arranged in a regular, repeating 3-D pattern called the SPACE LATTICE.
* Unit cell = smallest group of atoms which, repeated in 3-D, builds the whole crystal. Lattice parameters: a, b, c (edges) and angles alpha, beta, gamma.
* Amorphous solids (glass) have no long-range order. Most metals crystallise into BCC, FCC or HCP.
* Important terms: Coordination Number (CN) = number of nearest neighbour atoms; Atomic Packing Factor (APF) = volume of atoms in cell / volume of cell.

$$APF = (n \times 4/3\ \pi r^3) / V_{cell}$$

2. Crystallization of Metals

When a molten metal cools below its freezing point, solidification (crystallization) takes place in 2 steps:

* (i) NUCLEATION - formation of tiny stable solid nuclei in the liquid.
* (ii) GROWTH - atoms from the liquid attach to the nuclei, crystals grow until they meet each other.

Nucleation

* Homogeneous nucleation: nuclei form by themselves in pure liquid; needs large undercooling (delta T).
* Heterogeneous nucleation: nuclei form on impurities / mould walls; needs little undercooling. Used in practice (inoculants / grain refiners such as Ti, B in Al).
* Free energy change for a spherical nucleus of radius r :

$$\Delta G = -(4/3)\ \pi r^3 \Delta G_v + 4\ \pi r^2\ \gamma$$

* Volume term (negative) favours growth, surface-energy term (positive) opposes it. Maximum of Delta G occurs at the CRITICAL RADIUS r^*.

$$r^* = 2\ \gamma\ T_m / (\Delta H_f \cdot \Delta T)$$

* Embryo with $r < r^*$ dissolves; nucleus with $r > r^*$ is stable and grows. Greater

undercooling -> smaller r^* -> more nuclei -> FINE grains.

Cooling curve of a pure metal

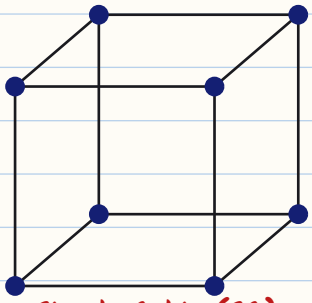
* Temperature falls, undercooling occurs slightly below freezing point, then rises back (latent heat released) and stays constant (thermal arrest) till solidification is complete; then falls again.

Growth - Dendritic growth

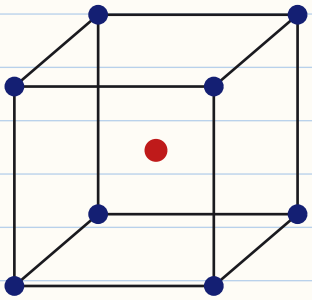
* Crystals grow fastest along certain directions, forming tree-like DENDRITES (primary arm with secondary and tertiary arms). Liquid between the arms finally freezes; the dendrites meet and form GRAINS separated by GRAIN BOUNDARIES.

Fast cooling -> more nuclei -> fine grains (stronger, tougher). Slow cooling -> few nuclei -> coarse grains (weaker).

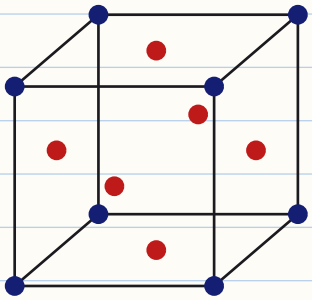
3. Crystal Structures and Packing Factor



Simple Cubic (SC)
n = 1, CN = 6



BCC
n = 2, CN = 8



FCC
n = 4, CN = 12

Property	SC	BCC	FCC	HCP
Atoms / cell (n)	1	2	4	6
Coord. number	6	8	12	12
a in terms of r	a = 2r	a = 4r/sqrt3	a = 4r/sqrt2	a = 2r
APF	0.52	0.68	0.74	0.74
Examples	Po	Fe(a),Cr,W,Mo	Al,Cu,Ni,Ag,Au	Mg,Zn,Ti,Cd

(a) Simple Cubic (SC)

* Atoms touch along the cube edge -> a = 2r. Atoms per cell = 8 x 1/8 = 1.

$$APF = (1 \times 4/3\ \pi r^3) / (2r)^3 = \pi/6 = 0.52$$

(b) Body Centred Cubic (BCC)

* Atoms touch along the body diagonal: sqrt3 a = 4r -> a = 4r/sqrt3. Atoms per cell = 8 x 1/8 + 1 = 2. CN = 8.

$$APF = (2 \times 4/3\ \pi\ r^3) / (4r/\sqrt{3})^3 = \sqrt{3}\ \pi / 8 = 0.68$$

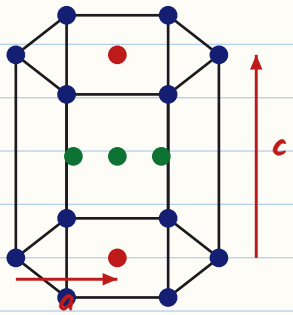
(c) Face Centred Cubic (FCC)

* Atoms touch along the face diagonal: sqrt2 a = 4r -> a = 4r/sqrt2 = 2 sqrt2 r. Atoms per cell = 8 x 1/8 + 6 x 1/2 = 4. CN = 12.

$$APF = (4 \times 4/3\ \pi\ r^3) / (2\sqrt{2}\ r)^3 = \pi / (3\sqrt{2}) = 0.74$$

* FCC has ABCABC... stacking of close packed planes {111}.

(d) Hexagonal Close Packed (HCP)



HCP unit cell

Atoms: 12 corners x 1/6 = 2

2 face-centre atoms x 1/2 = 1

3 inside atoms x 1 = 3 Total n = 6

CN = 12 , c/a = 1.633 (ideal)

* Stacking ABABAB... of close packed basal planes. a = 2r, ideal c/a = sqrt(8/3) = 1.633.
* Volume of HCP cell = (3 sqrt3 / 2) a^2 c.

$$APF = (6 \times 4/3\ \pi\ r^3) / ((3\sqrt{3}/2)\ a^2\ c) = \pi / (3\sqrt{2}) = 0.74$$

FCC and HCP are both CLOSE PACKED structures (APF = 0.74, CN = 12); only the stacking sequence differs.

4. Linear Density and Planar Density

$$LD = \frac{\text{no. of atoms centred on direction vector}}{\text{length of the vector}}$$

$$PD = \frac{\text{no. of atoms centred on the plane}}{\text{area of the plane}}$$

Linear density examples

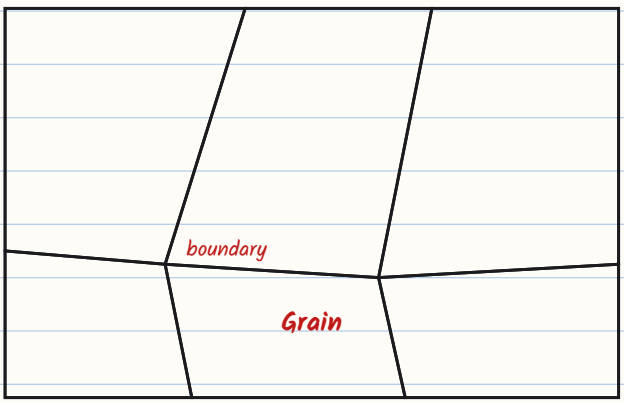
* SC [100]: 1 atom per edge length a -> LD = 1/a = 1/2r.
* BCC [111] (body diagonal): 2 atoms over length sqrt3 a = 4r -> LD = 2/(4r) = 1/2r (closest packed direction in BCC).
* FCC [110] (face diagonal): 2 atoms over length sqrt2 a = 4r -> LD = 2/4r = 1/2r (closest packed direction in FCC).
* FCC [100]: LD = 1/a = 1/(2 sqrt2 r) = 0.354/r.

Planar density examples

* SC (100) plane: 4 x 1/4 = 1 atom in area a^2 -> PD = $1/a^2$ = $1/4r^2$.
* BCC (110) plane: area = a x sqrt2 a; atoms = 4 x 1/4 + 1 = 2 -> PD = 2/(sqrt2 a^2) = 0.265/r^2 (most densely packed plane in BCC).
* FCC (111) plane: triangle of side sqrt2 a, atoms = 3 x 1/6 + 3 x 1/2 = 2 -> PD = 2 / ((sqrt3/2) a^2) = 4/(sqrt3 a^2) = 0.29/r^2 (closest packed plane of FCC, 6 neighbours in the plane).
* FCC (100): atoms = 4 x 1/4 + 1 = 2, area a^2 -> PD = $2/a^2$ = 0.25/r^2.

Slip occurs on planes of HIGHEST planar density along directions of HIGHEST linear density -> FCC metals (Cu, Al) are very ductile.

5. Grains, Grain Boundaries and Grain Size



Grain size (ASTM): N = 2^(n-1)

N = grains / sq. inch at 100x

n = ASTM grain size number

Larger n -> finer grains

Hall-Petch: sigma_y = sigma_0 + k d^(-1/2)

* GRAIN = single crystal region with uniform orientation. A metal piece (polycrystalline) is an aggregate of many grains.
* GRAIN BOUNDARY = narrow region (about 2-3 atom diameters) where neighbouring grains of different orientation meet; atoms are irregularly arranged, packing is poor.

Types

* Low angle (tilt) boundary: misorientation < 10 deg, made of array of edge dislocations. High angle boundary: misorientation > 15 deg.
* Twin boundary: mirror image across the boundary plane. Stacking fault: error in stacking sequence.

Effects of grain boundaries

* They block dislocation movement -> increase strength and hardness (Hall-Petch relation). Fine grain = high strength + toughness.
* They are high energy regions -> preferential etching (grain boundaries appear dark under the microscope), preferential corrosion, diffusion and precipitation.
* At high temperature (above 0.5 T_m) boundaries slide -> creep. Therefore coarse grained or single crystal blades are used in turbines.
* Boundaries scatter electrons -> slightly raise electrical resistivity.

$$\sigma_y = \sigma_0 + k_y d^{-1/2}$$

Determination of grain size

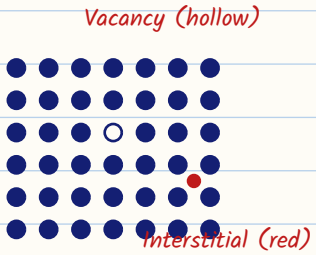
* ASTM method: sample polished, etched, examined at 100X magnification. Count grains per square inch N. Grain size number n :

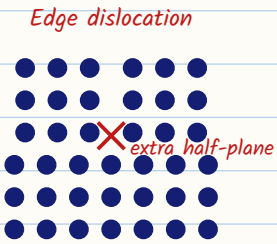
$$N = 2^{n-1}$$

* Comparison method: compare the micrograph with standard ASTM charts (n = 1 coarse to n = 8-10 fine).
* Intercept (Heyn) method: draw lines of known length L on the micrograph, count number of grain boundaries intersected P; mean intercept length = L/(P x M), M = magnification.
* Planimetric (Jeffries) method: draw a circle of known area, count grains inside (n_i) + half of grains cut by the circle ($n_c/2$); grains per mm^2 = f ($n_i + n_c/2$), f = Jeffries multiplier.

6. Imperfections in Crystals

Real crystals are never perfect. Imperfections (defects) control strength, ductility, conductivity and diffusion.





A. Point defects (0-dimensional)

* Vacancy: missing atom. Equilibrium concentration $n_v = N \exp(-Q_v/kT)$ (increases with temperature).
* Interstitial: extra atom squeezed into the space between lattice atoms (self interstitial or small impurity like C, N, H in iron).
* Substitutional impurity: foreign atom replaces a host atom.
* Frenkel defect: ion leaves its site and goes to interstitial position (vacancy + interstitial pair). Schottky defect: pair of cation + anion vacancies (ionic solids).

B. Line defects - Dislocations (1-dimensional)

* Edge dislocation: extra half plane of atoms inserted; dislocation line is the edge of this half plane. Burgers vector b is PERPENDICULAR to the dislocation line.
* Screw dislocation: crystal sheared so that atomic planes form a helical ramp. Burgers vector is PARALLEL to the line.
* Mixed dislocation: part edge, part screw.
* Burgers vector = magnitude and direction of lattice distortion (closure failure of Burgers circuit).
* Dislocation density = total length of dislocation line per unit volume (10^6 cm^{-2} annealed to 10^{12} cm^{-2} cold worked).

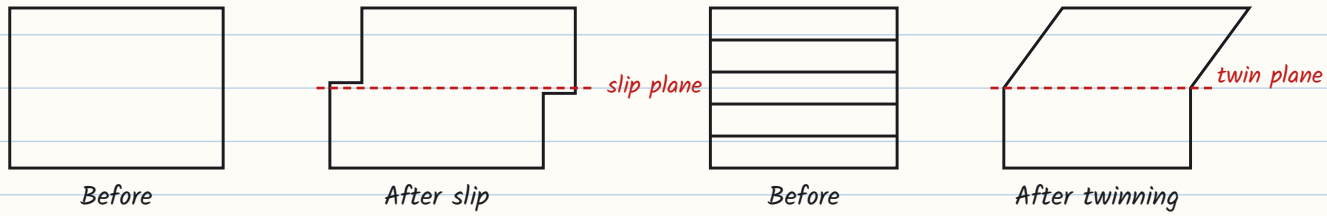
C. Surface (planar) defects (2-dimensional)

* External surface, grain boundaries, twin boundaries, stacking faults, phase boundaries.

D. Volume defects (3-dimensional)

* Pores, cracks, inclusions, voids, precipitates.

7. Slip and Twinning (Plastic deformation)



Slip

* Plastic deformation by sliding of one part of crystal over another along definite SLIP PLANES in definite SLIP DIRECTIONS. Slip plane + direction = SLIP SYSTEM.
* Slip occurs by movement of dislocations (needs much lower stress than shearing a perfect crystal - theoretical strength is about 1000 times larger than actual).
* Slip systems: FCC -> {111}<110> = 12 systems (very ductile). BCC -> {110}<111> (+{112},{123}) = 12 or more (fairly ductile). HCP -> basal (0001)<11-20> = only 3 at room temperature (brittle, e.g. Zn, Mg).
* Critical Resolved Shear Stress (Schmid's law): $\tau_R = \sigma \cos(\phi) \cos(\lambda)$.
* Slip lines are seen as fine steps on polished surface; a group of slip lines forms a slip band. Slip leaves the lattice orientation unchanged.

Twinning

* Part of the crystal shears so that the lattice becomes a MIRROR IMAGE of the parent across a TWIN PLANE. Atom movement is less than one atomic spacing and proportional to distance from the twin plane.
* Types: mechanical twins (by deformation, BCC and HCP, e.g. Fe at low T, Zn, Mg) and annealing twins (by recrystallisation in FCC like Cu, brass, Al).

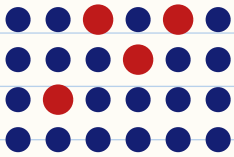
Slip	Twinning
Lattice orientation same on both sides	Orientation changes (mirror image)
Atoms move by whole lattice spacings	Movement < one atomic spacing
Takes place on widely spaced planes	Takes place on every plane in twinned region
Takes long time (ms)	Very fast (microseconds)
Needs lower stress, main mode	Needs higher stress, secondary mode

8. Constitution of Alloys

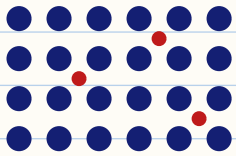
Necessity of alloying

* Pure metals are soft, weak, costly and have limited properties. ALLOY = metallic substance made of two or more elements, at least one a metal.
* Alloying is done to (i) increase strength, hardness, (ii) improve corrosion / oxidation resistance, (iii) improve casting, machining, weldability, (iv) get special magnetic / electrical / high temperature properties, (v) lower cost, (vi) allow heat treatment.
* Terms: Component = element or compound making the alloy. Phase = physically distinct, homogeneous, mechanically separable portion with its own boundary. System = group of alloys of same components. Microstructure = arrangement of phases seen under microscope.

Types of solid solutions



Substitutional



Interstitial

* SUBSTITUTIONAL solid solution: solute atoms replace solvent atoms at lattice sites (Cu-Ni, Cu-Zn). May be ordered or random.
* INTERSTITIAL solid solution: small solute atoms (H, C, N, B, O, radius < 0.59 x solvent) occupy voids between solvent atoms (C in Fe: ferrite, austenite).
* Terminal (primary) solid solution = based on one pure component; Intermediate solid solution = new structure, composition in the middle of diagram.

Hume-Rothery's rules (for extensive substitutional solid solubility)

* 1. ATOMIC SIZE FACTOR: difference in atomic radii must be less than 15%. Greater difference -> lattice strain, limited solubility.
* 2. CRYSTAL STRUCTURE: both elements should have the same crystal structure.
* 3. ELECTRO-NEGATIVITY (chemical affinity): small difference. Large difference -> compound formation instead of solution.
* 4. RELATIVE VALENCY: a metal of lower valency dissolves more easily in a metal of higher valency than vice-versa.

Cu-Ni satisfies all four rules (radius 0.128 / 0.125 nm, both FCC, similar electronegativity, valency +2) -> complete solubility.

Intermediate alloy phases

* Formed when two metals differ in chemical nature; the crystal structure is different from either component; usually hard and brittle; composition range narrow.
* (a) Interstitial compounds: transition metal + small atoms H, C, N, B (e.g. Fe3C, TiC, WC, Fe4N, TiN). Very hard. Hagg's rule: $r_x/r_M < 0.59$ gives simple structure.
* (b) Valency compounds (normal): follow chemical valency rules, e.g. Mg2Pb, Mg2Sn, Mg2Si, ZnS.
* (c) Size factor compounds: Laves phases AB2 (MgCu2, Mg2n2) - atomic radius ratio about 1.2; Sigma phase (Fe-Cr) - hard and brittle, harmful in stainless steels.

Electron compounds (Hume-Rothery phases)

* Intermediate phases whose crystal structure depends on the ratio of VALENCE ELECTRONS to ATOMS (e/a).

e/a ratio	Structure	Example
3/2 (21/14)	BCC (beta phase)	Cu2n, Cu5Sn, CuBe, Cu3Al
21/13	Complex cubic (gamma)	Cu52n8, Cu9Al4, Cu31Sn8
7/4 (21/12)	HCP (epsilon)	Cu2n3, Cu3Sn, Cu3Si

* Electron compounds are usually hard and brittle, with metallic bonding; they often have a wide composition range and show disorder-order transformation.

9. Equilibrium (Phase) Diagrams

* A phase diagram is a graph of TEMPERATURE vs COMPOSITION showing the phases present at equilibrium (very slow cooling). It is also called equilibrium / constitutional diagram.

Gibbs Phase rule

$$F = C - P + 2 \text{ (at constant pressure : } F = C - P + 1)$$

* F = degrees of freedom (variance), C = number of components, P = number of phases. For a binary alloy C = 2, so F = 3 - P. In a single phase region F = 2 (T and composition can be changed). Two-phase region F = 1. Three phase (invariant) point F = 0.

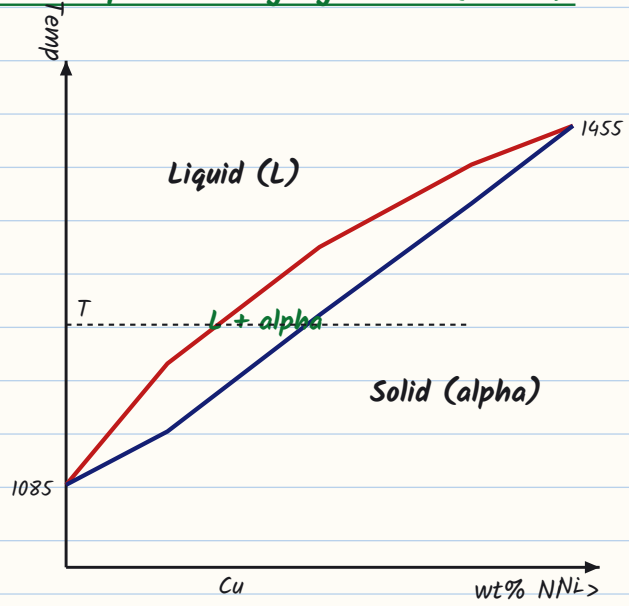
Experimental methods of construction

* 1. THERMAL ANALYSIS (cooling curves): a series of alloys is melted and cooled slowly, temperature noted with time. Change of slope / thermal arrest marks liquidus, solidus and

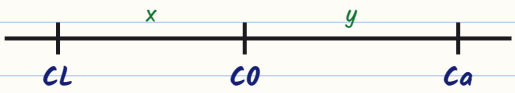
invariant temperatures. Most important method.

* 2. Microscopic examination: specimens quenched from various temperatures and the phases observed after polishing and etching.
* 3. X-ray diffraction: gives crystal structure and lattice parameter of the phases; solubility limits found from change of lattice parameter.
* 4. Dilatometry (change of length), electrical resistivity, magnetic and hardness measurements: detect solid state transformations.
* The points obtained from all alloys are plotted on T vs composition axes and joined: LIQUIDUS (start of freezing), SOLIDUS (end of freezing), solvus (solid solubility limit).

A. Isomorphous alloy systems (Cu-Ni)



Lever rule at temperature T :



W(liq) = (Ca - C0)/(Ca - CL)
W(alpha) = (C0 - CL)/(Ca - CL)
W(liq) + W(alpha) = 1

* Two metals completely soluble in BOTH liquid and solid states -> single solid phase alpha over whole range. Examples: Cu-Ni, Au-Ag, Ge-Si, Bi-Sb, Ni-Co.
* Three regions: Liquid (above liquidus), Liquid + alpha (between), alpha solid (below solidus). Melting points: Cu 1085 C, Ni 1455 C.

Equilibrium cooling of an alloy

* Alloy of composition C_0 cooled slowly: liquid cools till liquidus, first solid alpha (rich in Ni) forms. On further cooling, the composition of solid follows solidus and the liquid follows liquidus; the solid keeps changing composition by DIFFUSION so that at the solidus the entire alloy is uniform alpha of composition C_0. Heating is the reverse sequence.

Lever rule (to find amount of phases)

* Draw a horizontal TIE LINE at the temperature through the alloy composition. Its ends

give the compositions of the two phases. Amount of a phase = length of the lever ARM OPPOSITE to it / total tie-line length.

$$W_L = (C_a - C_0) / (C_a - C_L) \quad W_a = (C_0 - C_L) / (C_a - C_L)$$

Eg: 35 wt% Ni alloy at 1250 C: tie line gives C_L = 31.5% Ni and C_a = 42.5% Ni -> W_L = (42.5-35)/(42.5-31.5) = 0.68 (68% liquid) and W_a = (35-31.5)/(42.5-31.5) = 0.32 (32% solid).

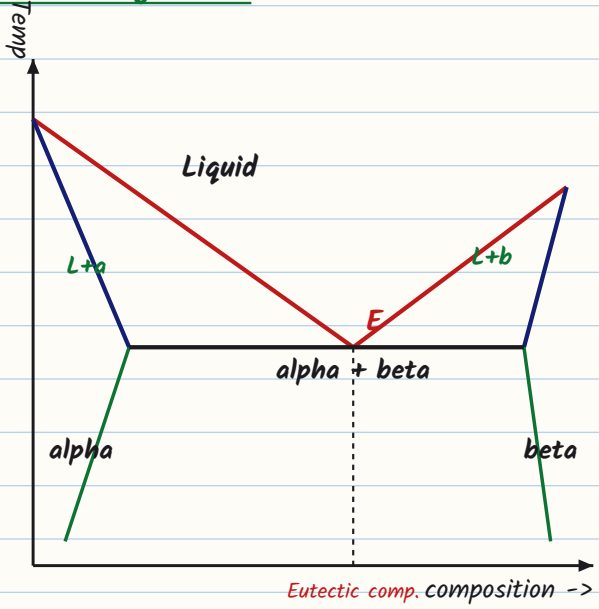
Coring (dendritic segregation)

* In NON-equilibrium (fast) cooling diffusion in the solid is too slow: centre of each grain is rich in the high melting component (Ni) and outer layers are rich in Cu. This cored structure causes variation in properties, lower melting point (hot shortness) and corrosion.
* Removed by HOMOGENIZING anneal (heating just below solidus for long time).

Miscibility gaps

* In some isomorphous systems, solid solution separates into two solid solutions of different composition at lower temperature (dome shaped region). Example: Au-Ni, Cu-Pt, Cr-W. Liquidus and solidus may show a minimum or maximum (azeotrope-like).

B. Eutectic systems



Eutectic reaction :

L (C_E) --cooling--> a + b

Invariant: F = 0 (3 phases in equilibrium, constant T)

Hypo-eutectic : C < C_E

Hyper-eutectic: C > C_E

Eg: Pb-Sn (61.9% Sn, 183 C)

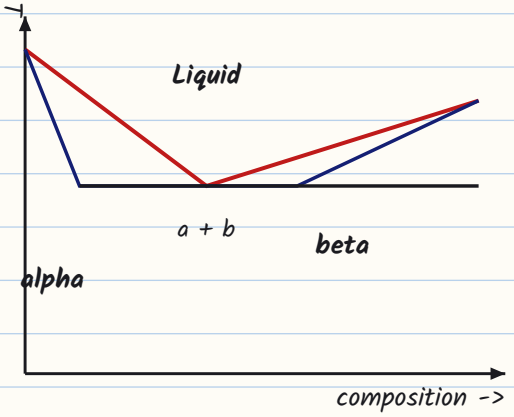
* Two components completely soluble in liquid but only PARTIALLY soluble in the solid state. Lowest melting alloy = EUTECTIC composition, which freezes at a single temperature like a pure metal.
* Eutectic alloy structure: fine mixture (lamellar) of alpha + beta. Hypoeutectic: primary alpha (proeutectic) + eutectic. Hypereutectic: primary beta + eutectic.

* Examples: Pb-Sn (solder), Al-Si, Cu-Ag, Pb-Sb (type metal), Fe-Fe3C (ledeburite).
* Solvus lines show decreasing solid solubility with fall in temperature -> basis of PRECIPITATION (age) hardening.

C. Congruent melting intermediate phases

* An intermediate compound that melts at a single temperature without change in composition (like a pure metal) is said to melt CONGRUENTLY, e.g. Mg2Pb, Mg2Sn in Mg-Pb. The diagram looks like TWO eutectic diagrams placed side by side with the compound's maximum in between.
* If the compound decomposes on heating into liquid of different composition + another solid, it melts INCONGRUENTLY (via a peritectic reaction).

D. Peritectic reaction



Peritectic reaction :

L + a (solid) ---> b (solid)

at constant T, F = 0

Eg: Fe-C at 1493 C: L + d -> g

Eutectoid : g -> a + Fe3C (727 C)

Peritectoid: a + b -> g (solid)

Monotectic: L1 -> L2 + solid

* Liquid + solid 1 ---> solid 2 on cooling at a constant temperature. Examples: Fe-C (1493 C), Cu-Zn, Pt-Ag, Fe-Ni.
* Peritectic reaction is slow; the new solid forms as a layer around the first solid, stopping diffusion -> a non-equilibrium (cored) structure usually results.

E. Transformations in the solid state

* ALLOTROPY (polymorphism): change of crystal structure in solid with temperature, e.g. Fe: BCC alpha (below 912 C) -> FCC gamma (912-1394 C) -> BCC delta (1394-1538 C). Also Ti (HCP -> BCC at 882 C), Sn (grey/white).
* EUTECTOID reaction: one solid phase -> two other solid phases on cooling. gamma (0.77% C) -> alpha + Fe3C (PEARLITE) at 727 C.
* PERITECTOID reaction: two solids -> one new solid on cooling: alpha + beta -> gamma. (Cu-Si, Ag-Al)
* Order-disorder transformation: e.g. CuZn (beta brass) above 460 C random, below it ordered.

Reaction	On cooling at constant T
Eutectic	Liquid -> solid 1 + solid 2
Eutectoid	Solid 1 -> solid 2 + solid 3
Peritectic	Liquid + solid 1 -> solid 2
Peritectoid	Solid 1 + solid 2 -> solid 3
Monotectic	Liquid 1 -> solid + liquid 2

Relationship between phase diagram and properties

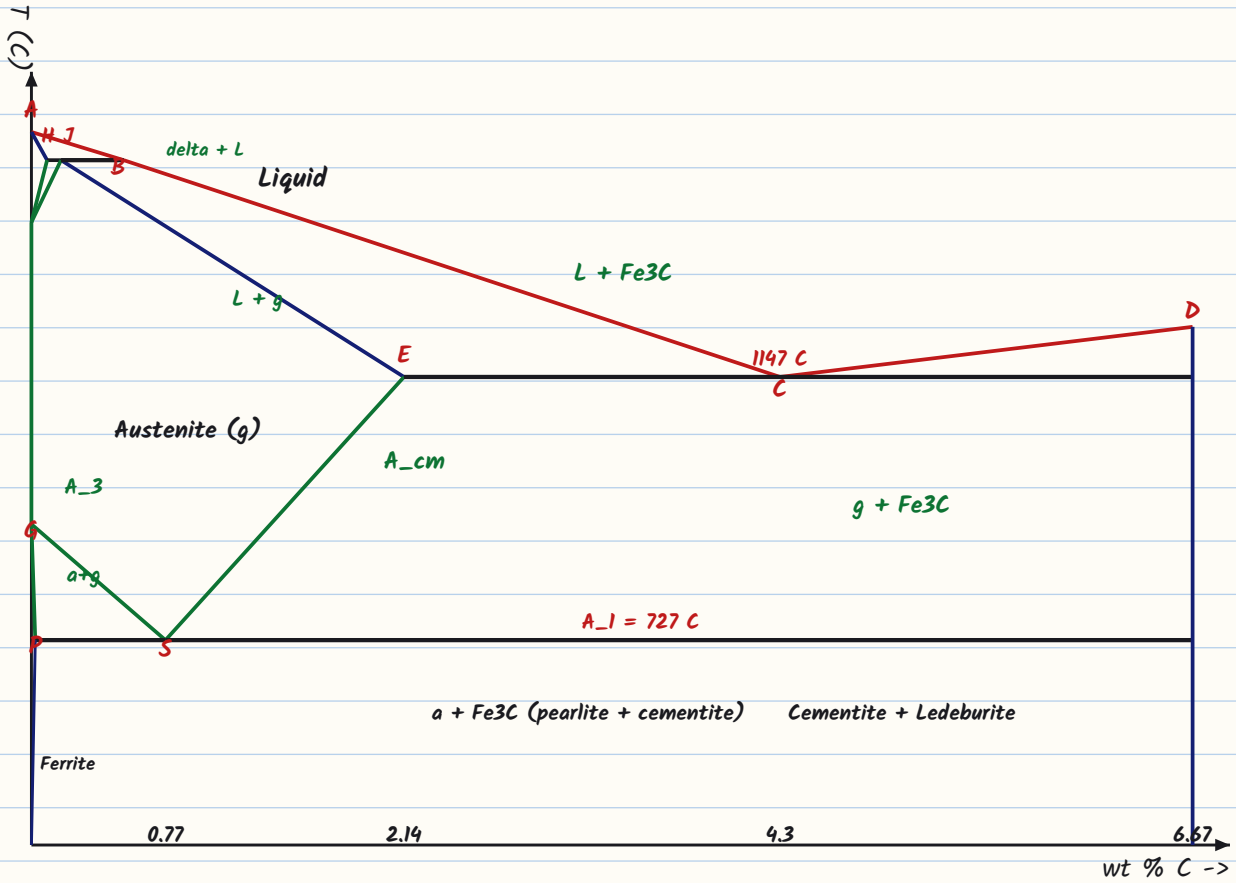
* Solid solution alloys (isomorphous): hardness / strength rise to a maximum near 50:50 composition; electrical conductivity falls to a minimum there (curved).
* Eutectic type (mechanical mixtures): properties vary LINEARLY with composition between the properties of the two phases.
* Compound formation: sharp peak (maximum hardness) at compound composition. Casting: eutectic alloys have the best fluidity and low shrinkage (good castability); wide freezing range alloys are poor in castability and prone to segregation.

10. Study of Binary Phase Diagrams

(i) Cu-Ni diagram - summary

* Isomorphous: single FCC alpha solid solution at all compositions. Liquidus and solidus lens shaped from 1085 C (Cu) to 1455 C (Ni).
* Properties: Cu-Ni alloys have excellent corrosion resistance and ductility; Monel (67 Ni - 30 Cu), Constantan (45 Ni - 55 Cu) used for resistors and thermocouples; cupro-nickel (70 Cu - 30 Ni) for marine condenser tubes and coins.
* Cannot be hardened by heat treatment, only by cold work (strain hardening) and solid solution strengthening.

(ii) Iron - Iron carbide (Fe-Fe3C) diagram



* Iron-carbon alloys up to 6.67% C (the compound Fe3C, CEMENTITE). Fe3C is metastable but treated as an equilibrium phase. Alloys < 2.14% C are STEELS; 2.14% - 6.67% C are CAST IRONS.

Phases

* FERRITE (alpha): interstitial solid solution of C in BCC iron; max 0.022% C at 727 C; soft, ductile, magnetic below 768 C (Curie temp.). delta-ferrite: BCC at high temperature (up to 0.09% C at 1493 C).
* AUSTENITE (gamma): interstitial solid solution of C in FCC iron; max 2.14% C at 1147 C; soft, ductile, non-magnetic; stable only above 727 C.
* CEMENTITE (Fe3C): intermetallic compound with 6.67% C, orthorhombic; very hard and brittle.
* PEARLITE: eutectoid mixture of alternate lamellae of ferrite + cementite (0.77% C), strong with fair ductility. LEDEBURITE: eutectic mixture of austenite + cementite (4.3% C, 1147 C); transformed ledeburite (pearlite + cementite) below 727 C.

Invariant reactions

Reaction	Temp.	Comp. (% C)	Equation
Peritectic	1493 C	0.17	L + delta-ferrite -> austenite
Eutectic	1147 C	4.3	L -> austenite + Fe3C
Eutectoid	727 C	0.77	austenite -> ferrite + Fe3C

Critical temperature lines

* A1 (727 C): eutectoid line, PSK. A3: GS line - ferrite begins to form from austenite (912 C for pure iron). Acm: SE line - cementite begins to separate from austenite in hypereutectoid steels. A2 : 768 C (magnetic change).

Microstructures on slow cooling

* EUTECTOID steel (0.77% C): all austenite -> 100% pearlite.
* HYPOEUTECTOID steel (< 0.77% C): proeutectoid FERRITE (white) + pearlite.
* HYPEREUTECTOID steel (0.77 - 2.14% C): proeutectoid CEMENTITE network along grain boundaries + pearlite.
* Cast irons: hypoeutectic (2.14-4.3% C): pearlite + cementite + transformed ledeburite; eutectic (4.3%): ledeburite; hypereutectic (4.3-6.67%): primary cementite + ledeburite.

Worked example (lever rule at just below 727 C)

Hypoeutectoid steel with 0.4% C:

% proeutectoid ferrite = (0.77 - 0.4)/(0.77 - 0.022) x 100 = 49.5%

% pearlite = (0.4 - 0.022)/(0.77 - 0.022) x 100 = 50.5%

Total ferrite (in ferrite + pearlite) = (6.67 - 0.4)/(6.67 - 0.022) = 94.3%, cementite = 5.7%.

Hypereutectoid 1.2% C steel: proeutectoid cementite = (1.2 - 0.77)/(6.67 - 0.77) = 7.3%; pearlite = 92.7%.

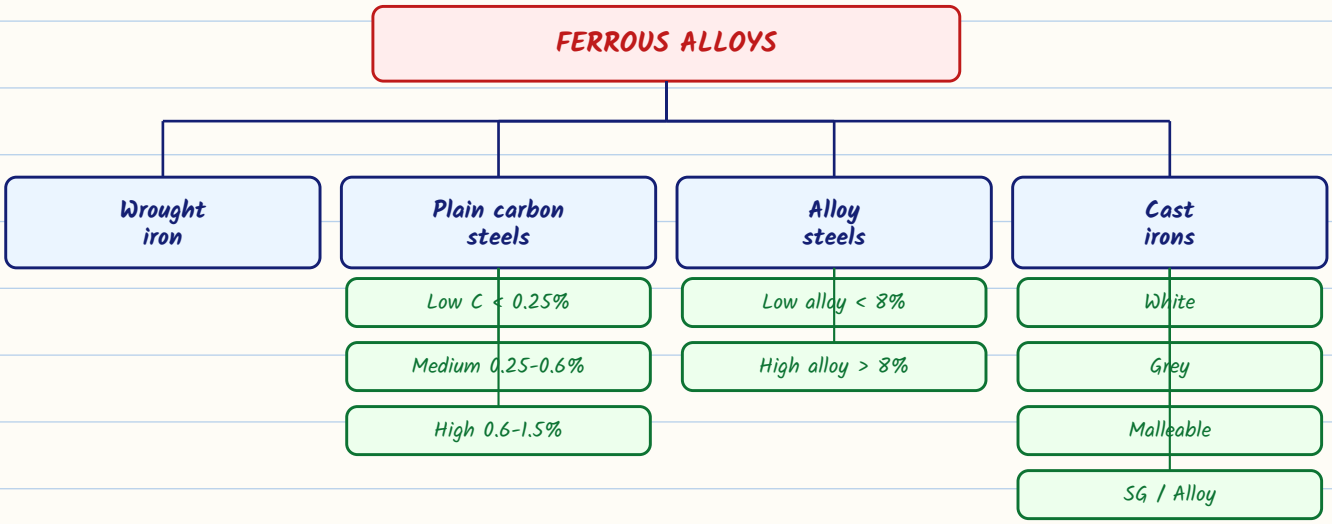
Quick revision

* APF: SC 0.52, BCC 0.68, FCC = HCP = 0.74. Slip systems: FCC 12, BCC 12+, HCP 3.
* Hume-Rothery rules: size < 15%, same structure, similar electronegativity, higher valency solvent.
* Phase rule F = C - P + 1. Invariant reactions: eutectic, peritectic, eutectoid, peritectoid, monotectic.
* Fe-Fe3C: 727 C eutectoid (0.77% C), 1147 C eutectic (4.3% C), 1493 C peritectic (0.17% C); steels < 2.14% C.

UNIT - II : FERROUS METALS AND ALLOYS

1. Introduction and classification

* Ferrous metals have IRON as the base: steels and cast irons. Iron is the cheapest and most used engineering metal - about 90% of all metal tonnage.
* Carbon content decides the family: Wrought iron < 0.08% C, Steels 0.08 - 2.14% C, Cast irons 2.14 - 6.67% C (usually 2.5 - 4% C with 1 - 3% Si).



2. Cast Irons

* Cast iron = Fe - C - Si alloy with C above 2.14% (in practice 2.5 - 4.0%), Si 1 - 3%, Mn, S, P in small amounts. Good fluidity, low melting point (1150 - 1300 C), easy casting, excellent damping, wear resistance and machinability; but low ductility.
* Carbon appears either as CEMENTITE (Fe3C - white iron) or as free

GRAPHITE (grey, malleable, SG). Factors: cooling rate, composition (Si, a graphitiser), section thickness.

Si promotes graphite; S, Mn, Cr and fast cooling promote cementite (white iron). 'Greener' graphite -> softer, more machinable iron.

(a) White cast iron

* All carbon is combined as Fe3C. Produced by rapid cooling (chilling) of low Si (0.5 - 1.3%) melt; fracture is bright white and crystalline.
* Structure: pearlite + cementite + (transformed) ledeburite. Very hard (350-500 BHN), very brittle, highly wear and abrasion resistant, not machinable (only grinding), tensile strength about 200-400 MPa, compression strong.
* Uses: crushing and grinding balls, rolls, liners, ploughshares, rail brake shoes, extrusion nozzles, and as the parent material for malleable iron. Chilled iron castings have white iron surface over grey core (car wheels, rolls, cams).

(b) Grey cast iron

* Composition: 2.5-4% C, 1-3% Si, 0.4-1% Mn, S < 0.15%, P < 0.3%. Carbon is present as FLAKES of graphite in a matrix of ferrite, pearlite or both. Fracture appears grey.
* Slow cooling and high Si produce graphite flakes which act as internal notches -> low tensile strength (150 - 400 MPa, grades FG 150 to FG 400), almost no ductility (< 1%), but compressive strength 3 - 4 times the tensile.
* Properties: excellent machinability, castability, wear resistance (graphite is a solid lubricant), very good vibration DAMPING and resistance to compressive loads; low cost; poor impact strength.

* Uses: machine tool beds, engine blocks, cylinder heads and liners, gear housings, pipes, pulleys, brake drums, manhole covers, flywheels.

(c) Malleable cast iron

* White iron (C 2.2-2.9%, Si 0.9-1.9%) given long MALLEABILISING heat treatment (annealing) which breaks down cementite: Fe3C -> 3Fe + C (temper carbon, compact rosettes).

Type	Process	Structure	Property
Whiteheart	Heat 900-1050 C in oxidising medium (haematite ore), 60-100 h	Ferrite + temper carbon, decarburised skin	Less ductile, weldable
Blackheart	Stage I 900-950 C (48 h) + slow cool to 700 C, neutral atm.	Ferrite matrix + nodules of temper carbon	Ductile, good machinability
Pearlitic	Cool faster, then 2nd stage short	Pearlite + temper carbon	Stronger, wear resistant

* Tensile strength 300-500 MPa, elongation 6 - 14%, good shock resistance and machinability. Uses: automobile parts (crankshafts, connecting rods, differential housings), pipe fittings, hand tools, agricultural implements, railroad castings. Limited to thin sections (< 40 mm).

(d) Spheroidal graphite (SG / ductile / nodular) cast iron

* Graphite appears as SPHEROIDS (nodules). Obtained by adding 0.03 - 0.06% MAGNESIUM (or Ce) to the melt (after desulphurising) followed by inoculation with ferro-silicon (0.5-1%).
* Sphere shape removes stress concentration -> high strength (400 - 800 MPa) AND ductility (2 - 18%). Matrix may be ferritic (ductile),

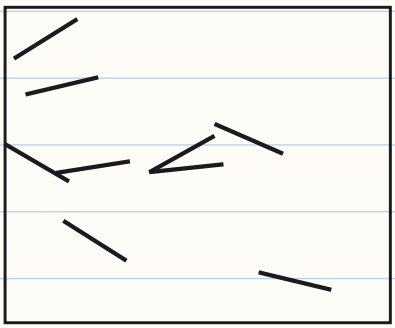
pearlitic (strong) or bainitic / martensitic (ADI: austempered ductile iron).

* Uses: crankshafts, pistons, gears, pipes (water, sewage), dies, valves, hydraulic parts, wind turbine hubs, automobile suspension parts, steel mill rolls.

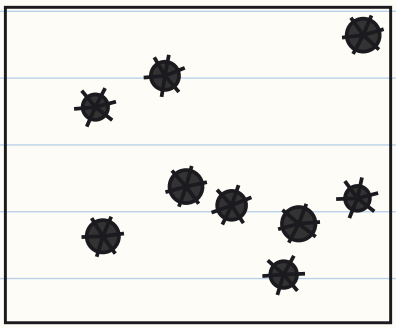
(e) Alloy cast irons

* Contain special alloying elements to give high strength, wear, heat and corrosion resistance.
* Ni-hard (4% Ni, 2% Cr): martensitic white iron for abrasion. Ni-resist (14-30% Ni): austenitic, corrosion and heat resistant. Silal / high-silicon (14-17% Si): acid resistant (Duriron). High Cr iron (12-35% Cr): heat and abrasion resistant (pump casings, furnace parts). Mo and Cu increase strength of grey iron.

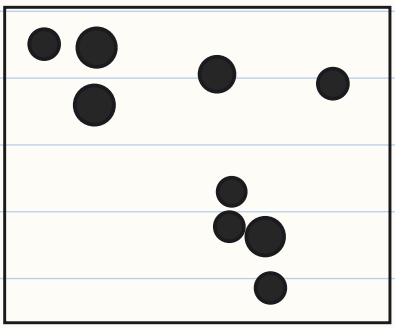
Property	White	Grey	Malleable	SG iron
Graphite	None	Flakes	Rosettes	Spheres
Tensile MPa	200-400	150-400	300-500	400-800
Ductility %	~0	< 1	6-14	2-18
Machinability	Poor	Excellent	Good	Good



Grey iron: FLAKES



Malleable: ROSETTES



SG iron: SPHEROIDS

Effect of elements on cast iron

* CARBON: increases graphite, lowers melting point and strength.

SILICON: strong graphitiser (2-3%), improves fluidity and acid resistance. MANGANESE: opposes graphitisation (carbide former), neutralises S, adds strength (0.5-1%).

* SULPHUR: carbide stabiliser, causes hot shortness, reduces fluidity (keep < 0.1%). PHOSPHORUS: forms hard, brittle steadite (Fe-Fe3C-Fe3P eutectic, m.p. 950 C), increases fluidity (thin castings, ornamental).
* Carbon equivalent CE = %C + (%Si + %P)/3 ; CE = 4.3 is eutectic composition. Higher CE gives lower strength grey iron.

Cast iron vs steel

Cast iron	Steel
2.14 - 6.67% C, Si 1-3%	0.05 - 2.14% C, Si < 0.6%
Cast only (brittle), hardly forgeable	Can be rolled, forged, welded, heat treated
Low cost, damping, wear resistance	Higher tensile strength and toughness
Melting point 1150-1300 C	Melting point 1370-1530 C

3. Steels

Classification of steels

* By CARBON: Low carbon / mild steel (0.05 - 0.25% C), medium carbon (0.25 - 0.6%), high carbon (0.6 - 1.5%).
* By ALLOYING: plain carbon steels (only C, Mn <= 1.65%, Si <= 0.6%), low alloy (total alloy < 8%), high alloy (> 8%: stainless, tool steels).
* By DEOXIDATION: killed, semi-killed, rimmed steel. By application: structural, tool, spring, stainless, high speed.

* By structure: ferritic, pearlitic, austenitic, martensitic, bainitic. IS designation: 40C8 (0.40% C, 0.8% Mn), 20CrlMo28.

Plain carbon steels – structure and properties

* As carbon rises: hardness, tensile strength, wear resistance increase; ductility, toughness, weldability and machinability decrease. Structure follows Fe-Fe3C diagram: ferrite + pearlite (hypo), pearlite (0.77%), pearlite + Fe3C network (hyper).

Type	%C	Structure	Properties	Uses
Dead mild	0.05-0.15	Mostly ferrite	Very ductile, weldable, UTS 300-400	Wires, sheets, rivets, pipes, car bodies
Mild	0.15-0.3	Ferrite + little pearlite	UTS 400-500, tough	Shafts, bolts, structural, case hardening
Medium	0.3-0.6	Ferrite + pearlite	UTS 550-800, heat treatable	Axles, gears, crankshafts, rails
High	0.6-1.0	Mostly pearlite	Hard, strong, less ductile	Springs, hammers, dies, wire ropes
Very high	1.0-1.5	Pearlite + Fe3C	Very hard, wear resist., brittle	Files, razors, drills, cutters

* Effect of impurities: Mn (deoxidiser, forms MnS, strength); Si (deoxidiser, strength); S (hot shortness - forms FeS, so Mn added); P (cold shortness - brittleness, but improves machinability); O, N, H harmful.

Low alloy steels

* Alloy additions < 8% (Ni, Cr, Mo, V, Mn, Si, W) to increase hardenability, strength, toughness, and low temperature / corrosion

properties without losing weldability.

* HSLA (high strength low alloy) steels: < 0.2% C with Cu, V, Nb, Ti (micro-alloys); yield 300-550 MPa; used in bridges, pipelines, car frames, pressure vessels.
* Examples: 1-3% Ni steel (toughness; gears), Ni-Cr steel (shafts, gears), Cr-Mo steels (pressure vessels, high temp.), Cr-V steel (springs, tools), Si-Mn steel (springs - 60Si2Mn).

Hadfield manganese steel

* Composition: 1.0 - 1.4% C and 10 - 14% Mn (about 12% Mn) - AUSTENITIC steel. After water quenching from 1000 - 1100 C (toughening treatment) the structure is fully tough austenite.
* Special property: WORK HARDENING - the surface becomes extremely hard (500+ BHN) under impact or abrasion while the core remains tough and ductile. Non-magnetic, very poor machinability (use grinding), difficult to weld.
* Uses: rock crusher jaws, rail-road points and crossings, excavator bucket teeth, dredger parts, grinding mill liners, bullet-proof plates, safes.

Effect of alloying elements in steels

Element	Effect on steel
Nickel (Ni)	Stabilises austenite, increases toughness, strength and low-temperature impact; corrosion resistance
Chromium (Cr)	Carbide former, hardenability, wear, corrosion and oxidation resistance; stabilises ferrite
Molybdenum (Mo)	Hardenability, high temperature strength, creep resistance; reduces temper brittleness
Vanadium (V)	Strong carbide former, fine grain, secondary hardening, toughness
Tungsten (W)	Hot hardness and red hardness, wear resistance (HSS)
Manganese (Mn)	Hardenability, strength; at 12% gives work hardening austenite
Silicon (Si)	Deoxidiser, strengthens ferrite, magnetic and spring properties
Cobalt (Co)	Increases red hardness; permanent magnets
Titanium / Niobium	Strong carbide formers; stabilise stainless steel against intergranular corrosion

Stainless steels

* Minimum 10.5 - 12% Cr forms thin passive Cr2O3 film. Ferritic (12-27% Cr, magnetic), Martensitic (12-18% Cr, 0.1-1% C, hardenable: cutlery, turbine blades), Austenitic (18% Cr - 8% Ni, '18/8', non-magnetic, excellent corrosion resistance: kitchenware, chemical plant).

Type	Typical grade	Structure	Remarks
Ferritic	430: 17Cr, 0.1C	BCC ferrite, magnetic	Not hardenable; decorative, automobile trim
Martensitic	410: 12Cr, 0.15C; 420: 13Cr, 0.35C	Martensite after quench	Hardenable; cutlery, turbine blades, surgical tools
Austenitic	304: 18Cr-8Ni; 316: + 2-3% Mo	FCC, non magnetic	Best corrosion, weldable; not hardenable by heat treatment
Precipitation hardening	17-4 PH, 17-7 PH	Martensite + Cu or Al ppts	High strength aerospace
Duplex	2205: 22Cr-5Ni	Ferrite + austenite	Stress corrosion resistant

* Sensitisation / weld decay: in 18/8 steel heated at 500-800 C chromium carbide ($Cr_{23}C_6$) forms at grain boundaries, depleting Cr -> intergranular corrosion. Prevented by low C (304L), stabilising with Ti / Nb (321, 347) or solution annealing at 1050 C.

Other special steels

* FREE CUTTING steels: added S (0.1-0.3%), Pb or P -> easy chip breaking, used for screws, bolts made on automatic machines (e.g. 11SMn30).
* SPRING steels: 0.5-1.0% C with Si-Mn or Cr-V; high elastic limit and fatigue strength (leaf and coil springs; 55Si2Mn90, 50CrIV23).
* MARAGING steels: 18% Ni, Co, Mo, Ti with very low C; martensite + age hardening at 480 C gives UTS up to 2400 MPa and good toughness (rocket motor cases, dies).
* HEAT RESISTING steels: high Cr, Ni, Si (e.g. 25Cr-20Ni) for furnace parts; BEARING steel: 1% C - 1.5% Cr (EN31) for ball and roller

bearings; electrical (Si) steel for transformers.

Tool and die steels

* Used for making cutting tools, dies and punches; need high hardness (60-67 HRC), wear resistance, red hardness (hot hardness), toughness and dimensional stability. Always heat treated (harden + temper).

Class	Typical composition	Properties and uses
Plain carbon tool steel	0.6-1.5% C	Water hardening, loses hardness above 250 C; files, chisels, punches
High speed steel (HSS) 18-4-1	0.7C - 18W - 4Cr - 1V	Red hardness to 600 C; drills, milling cutters, lathe tools
Mo-HSS 6-6-4-2	6W-5Mo-4Cr-2V	Cheaper, tougher HSS; taps, reamers
Hot work die steel (H-series)	5% Cr, Mo, V, W	Resist thermal fatigue; forging / extrusion / die casting dies
Cold work (D-series)	1.5C - 12Cr - Mo	High wear, minimum distortion; blanking dies, gauges
Shock resisting (S)	0.5C - Si - Cr - W	Toughness; chisels, rivet sets, punches

Rule: Cutting tool materials in order of hot hardness: carbon steel < HSS < cemented carbides < ceramics < diamond.

NON-FERROUS METALS AND ALLOYS

* All metals other than iron-base: lighter, better corrosion resistance, high electrical / thermal conductivity, easy fabrication, special

properties; costlier than steel.

4. Copper and its alloys

* Pure Cu: FCC, density 8.96 g/cc, m.p. 1085 C, very high electrical (second to Ag) and thermal conductivity, ductile, corrosion resistant, non-magnetic. Uses: wires, cables, busbars, heat exchangers, roofing. Types: ETP copper (99.9%, 0.04% O), OFHC (oxygen free high conductivity) and deoxidised copper.

Alloy	Composition	Properties	Uses
Cartridge brass	70 Cu - 30 Zn	Alpha brass; best ductility, deep drawing	Cartridge cases, radiators
Muntz metal	60 Cu - 40 Zn	Alpha + beta; hot workable, strong	Condenser tubes, bolts, ship sheathing
Gilding metal	95 Cu - 5 Zn	Colour of gold	Jewellery, coins, medals
Admiralty brass	70 Cu-29 Zn-1 Sn	Sea water resistant	Condenser tubes
Tin bronze (gun metal)	88 Cu-10 Sn-2 Zn	Strong, wear resistant	Bearings, gears, valves, guns
Phosphor bronze	Cu-4-10 Sn-0.3 P	Fatigue resistant, spring	Springs, bearings, gears
Aluminium bronze	Cu-5-11 Al	High strength, corrosion	Propellers, pump parts
Beryllium bronze	Cu-2 Be	Age hardenable, 1400 MPa	Springs, non-sparking tools
Cupro-nickel / Monel	Cu-30 Ni	Corrosion resistance	Coins, condenser tubes

* BRASS = Cu + Zn (up to 40-45%), BRONZE = Cu + Sn (or Al, Si, Be). Alpha brass (< 36% Zn): single phase FCC, cold workable; alpha + beta brass (36 - 45% Zn): stronger but harder to cold work. Season cracking and dezincification are typical defects of brass.

5. Aluminium and its alloys

* Pure Al: FCC, density 2.7 g/cc (1/3 of steel), m.p. 660 C, excellent corrosion resistance (Al2O3 film), electrical conductivity 62% of Cu, very ductile, non-magnetic, non-toxic, easily recyclable. Low strength (UTS ~ 90 MPa) - improved by alloying, cold work and age hardening.
* Wrought alloys (designation 1xxx to 8xxx): 1xxx pure Al, 2xxx Cu (duralumin), 3xxx Mn, 4xxx Si, 5xxx Mg, 6xxx Mg + Si, 7xxx Zn (highest strength), 8xxx others.

Alloy	Composition	Notes / Uses
Duralumin (2017)	4 Cu - 0.5 Mg - 0.7 Mn - Al	Age hardenable; aircraft structure, rivets, automobile
Y-alloy	4 Cu - 2 Ni - 1.5 Mg	Retains strength at high T; pistons, cylinder heads
Hindalium	Al-Mg-Mn-Cr	Anodised; utensils
Magnalium	2-10 Mg - Al	Light weight; aircraft parts, balances
Al-Si alloy (Silumin)	10-13 Si	Eutectic; excellent castability: pistons, engine blocks
7075	5.6 Zn-2.5 Mg-1.6 Cu	UTS up to 570 MPa; aircraft wings
6061	Mg + Si	Weldable, extruded sections, frames

* Age hardening of duralumin: solution treat at 500 C, quench, age at room temp. (natural) or 150 C (artificial) -> GP zones / theta'

(CuAl2) precipitates.

6. Titanium and its alloys

* Pure Ti: HCP alpha below 882 C, BCC beta above; density 4.5 g/cc, m.p. 1668 C, high strength-to-weight ratio, outstanding corrosion resistance (TiO2 film), low thermal expansion; costly (Kroll process); poor machinability; reacts with O, N, H at high T so melted in vacuum / argon.
* ALPHA alloys (Ti-5Al-2.5Sn): weldable, creep resistant, not heat treatable. BETA alloys (Ti-13V-11Cr-3Al): hardenable, formable. ALPHA-BETA alloys (Ti-6Al-4V, 'workhorse', 50% of all Ti use): heat treatable, UTS ~ 1000 MPa.
* Uses: aircraft engine compressor blades, airframes, missile parts, chemical plant, surgical implants (bio-compatible), hip joints, marine parts, sports goods.

7. Magnesium and its alloys

* Mg: HCP, density 1.74 g/cc (lightest structural metal), m.p. 650 C, high damping, low modulus, poor corrosion and fatigue resistance, burns in air if finely divided, poor formability at room temperature (HCP: only 3 slip systems) - worked hot.
* Alloys: Mg-Al-Zn (AZ31 wrought, AZ91 die casting), Mg-Zn-Zr (ZK60), Mg-Th (high temperature), Elektron (Mg-Al-Zn), Mg-Li (ultra light). Uses: aircraft and automobile parts (wheels, gear box housings), laptops and phone bodies, portable tools, reflectors, sacrificial anodes.

8. Super alloys

* Alloys (Ni, Co or Fe base) with excellent strength, creep, fatigue, oxidation and corrosion resistance at VERY HIGH temperatures (up to

0.8-0.9 T_m, > 1000 C). They are for gas turbine and jet engine parts.

* Nickel base (Inconel, Nimonic, Waspaloy, Udimet, Hastelloy): matrix gamma (FCC Ni solid solution) strengthened by gamma-prime precipitates Ni3(Al,Ti) (coherent) + carbides (M23C6, MC) at grain boundaries.
* Cobalt base (Stellite, Haynes 25, Vitallium): solid solution + carbide strengthening; good hot corrosion and wear resistance; used in vanes, bio-implants.
* Iron-nickel base (A-286, Incoloy 800): cheaper, austenitic stainless type + gamma-prime. Strengthening mechanisms: solid solution, precipitation, dispersion (ODS - oxide dispersion strengthened), directional solidification and SINGLE CRYSTAL blades (removing grain boundaries stops creep).
* Uses: turbine blades and discs, combustion chambers, afterburners, nuclear reactor parts, heat exchangers, rocket engine nozzles, chemical reactors.

Comparison of light metals

Metal	Density g/cc	M.P. C	Structure	UTS (alloyed) MPa	Key use
Aluminium	2.7	660	FCC	200-570	Aircraft, packaging, cables
Magnesium	1.74	650	HCP	200-350	Lightweight casings
Titanium	4.5	1668	HCP/BCC	900-1200	Jet engines, implants
Copper	8.96	1085	FCC	250-1400 (Be-Cu)	Electrical, bearings
Nickel SA	8.2-8.9	1350	FCC	1000-1400	Turbine blades

Important points for exams

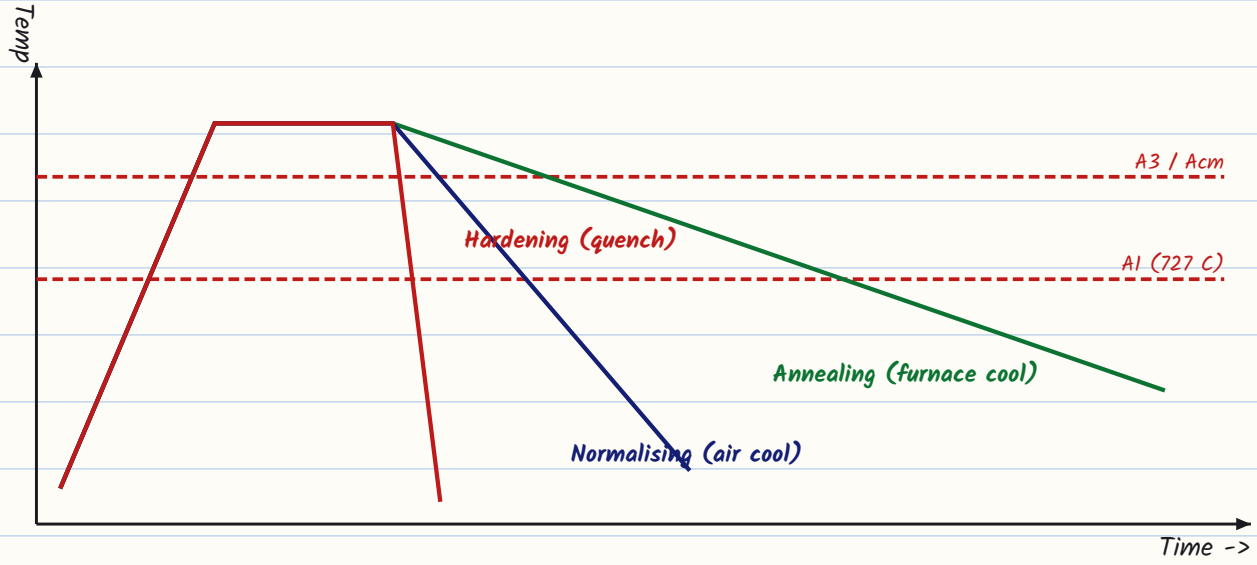
* 1. Distinguish white, grey, malleable and SG cast irons (graphite form, process, properties, uses).
* 2. Classify steels; explain properties of plain carbon steel in relation to % C.
* 3. Hadfield manganese steel: composition, toughening heat treatment, work hardening, uses.
* 4. Compositions and applications of brass, bronze, duralumin, Ti-6Al-4V, Mg alloys and super alloys.
* 5. Why are super alloys used in gas turbines? (gamma-prime strengthening, creep resistance, oxidation resistance, single crystal blades).

Revision: Hadfield steel = 12% Mn work hardening; Nimonic = Ni-Cr-Ti-Al super alloy; Duralumin = Al-Cu age hardening; Ti-6Al-4V = aerospace; SG iron = Mg added; Malleable iron = annealed white iron.

UNIT - III : HEAT TREATMENT OF STEELS

1. Introduction

* HEAT TREATMENT = controlled heating and cooling of a metal in solid state to obtain desired structure and properties (hardness, strength, toughness, ductility, machinability, stress relief).
* Any heat treatment has 3 steps: (i) HEATING to a definite temperature at a suitable rate, (ii) SOAKING (holding) for sufficient time for uniform temperature and structure change, (iii) COOLING at a definite rate (furnace, air, oil, water, brine).
* Soaking time: about 1 hour per 25 mm thickness (+ heat up time). Heating atmosphere: oxidising / reducing / neutral / vacuum / salt bath to avoid scaling and decarburisation.



2. Effect of Alloying Elements on Fe - Fe3C System

* Alloying elements shift the eutectoid TEMPERATURE and eutectoid COMPOSITION (% C) and change the austenite phase field.

* AUSTENITE STABILISERS (expand gamma field, lower A1 and A3): Ni, Mn, Co, Cu, C, N. With enough Ni (> 8%) / Mn (12%) austenite is retained at room temperature (austenitic stainless steel, Hadfield steel).
* FERRITE STABILISERS (shrink gamma loop, raise A1): Cr, Si, Mo, W, V, Al, Ti, Nb. With > 13% Cr the gamma field vanishes completely - ferritic stainless steels.
* Eutectoid carbon content is LOWERED by almost all alloying elements (Ti, Mo, W, Si, Cr strongly): e.g. at 5% Cr, eutectoid is at about 0.5% C; the eutectoid temperature is raised by Cr, W, Mo, Si and lowered by Ni, Mn.

Element	A1 temp	Eutectoid % C	Type
Ni	Lowers	Lowers	Austenite former
Mn	Lowers	Lowers	Austenite former
Cr	Raises	Lowers	Ferrite former + carbide
Mo, W, V	Raises	Lowers	Ferrite former + strong carbide
Si	Raises	Lowers	Ferrite former, graphitiser

* Carbide formers (Cr, W, Mo, V, Ti) produce alloy carbides (Cr_7C_3, WC, Mo_2C, VC) -> hardness, wear resistance and secondary hardening; most alloying elements (except Co) SHIFT THE TTT CURVE TO THE RIGHT (delayed transformation) -> improved hardenability.

3. Annealing

* Heating steel above (or around) critical temperature, soaking and cooling SLOWLY (in furnace) to obtain a soft, ductile, stress free,

coarse pearlite structure.

* Purposes: soften steel for machining or cold working, relieve internal stresses, refine grain, remove gases, improve ductility, homogenise structure, prepare for further heat treatment.

Types of annealing

* FULL ANNEALING: hypoeutectoid steel heated to 30 - 50 C ABOVE A3 (hypereutectoid: above A1), soaked, cooled in furnace slowly (10-30 C/h). Structure: coarse pearlite + ferrite (or cementite).
* PROCESS (sub-critical) ANNEALING / recrystallisation anneal: heating cold worked low carbon steel to 550-650 C (below A1), restores ductility by recrystallisation.
* STRESS RELIEF annealing: 500 - 650 C for welded / cast / machined parts, slow cool. No structure change.
* SPHEROIDISING annealing: high carbon steel (> 0.6% C) held just below A1 (650-700 C) for long hours (or cycled about A1) -> cementite lamellae turn into SPHEROIDS in ferrite; best machinability, maximum softness (spheroidite).
* DIFFUSION / HOMOGENISING annealing: 1000 - 1200 C for long time to remove coring and segregation in ingots and castings.

4. Normalizing

* Heating 30 - 50 C ABOVE A3 (for hypoeutectoid) or above Acm (hypereutectoid), soaking, then cooling in STILL AIR.
* Faster cooling than annealing gives FINE pearlite with finer grains, more strength and hardness (about 15-20% more than annealed), good toughness, uniform structure.

Annealing	Normalising
Furnace cooling (slow)	Air cooling (faster)
Coarse pearlite; softest	Fine pearlite; harder, stronger
Costly, time consuming	Cheaper, quicker
For better machinability, cold working	To refine grain, remove casting / forging stresses, improve strength
Lower strength, high ductility	Higher strength, good toughness

* Used for castings, forgings, welded parts and before hardening to get a uniform structure.

5. Hardening

* Heat to 30 - 50 C above A3 (hypoeutectoid) or above A1 (hypereutectoid, NOT above Acm), soak till all pearlite changes to austenite, then QUENCH rapidly (faster than critical cooling rate) in water, brine, oil, or polymer to get MARTENSITE.
* MARTENSITE: super-saturated interstitial solid solution of carbon in BCT (body centred tetragonal) iron, needle / lath shaped; formed by diffusionless shear transformation of austenite between Ms and Mf temperatures; very hard (up to 65 HRC), strong and brittle.

Austenite (FCC) --rapid quench--> Martensite (BCT)

* Hardness of martensite depends on CARBON content only (0.6% C gives about 65 HRC). Ms (C) = 539 - 423 (%C) - 30.4 (%Mn) - 17.7 (%Ni) - 12.1 (%Cr) - 7.5 (%Mo) degrees C. Above 0.5% C a part of austenite remains as RETAINED AUSTENITE.
* Quenching media in order of severity: Brine > Water > Mineral oil >

Polymer > Air > Furnace. Quench stages: vapour blanket -> boiling -> convection.

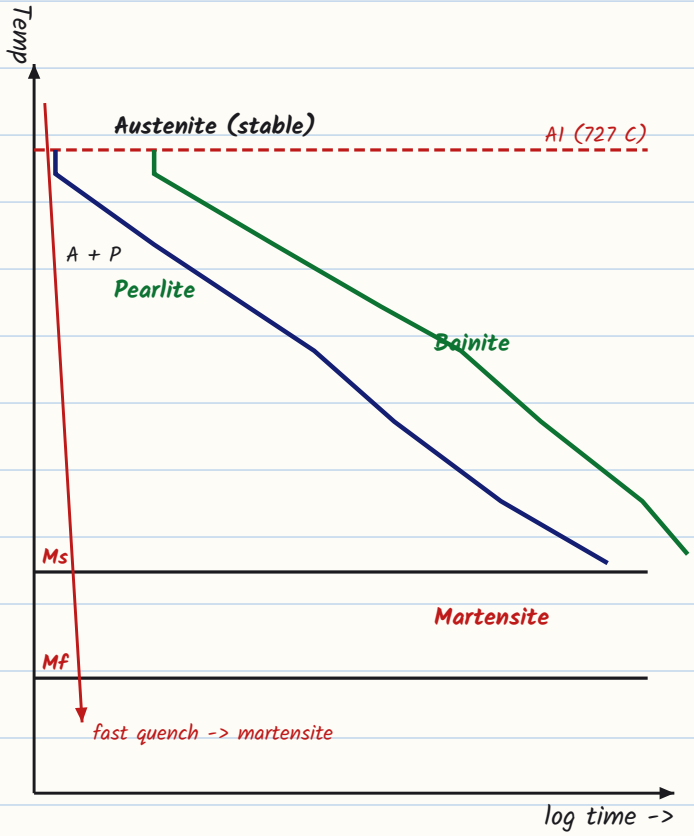
* Defects: quench cracks, distortion, soft spots, residual stresses, decarburisation. Remedies: use alloy steel with oil quench, interrupted quench, tempering immediately.

Critical cooling rate (CCR)

* The minimum cooling rate which just avoids the pearlite / bainite 'nose' of the TTT curve and gives fully martensitic structure. Alloying (except Co) lowers CCR -> easier to harden.

6. TTT Diagrams (Isothermal Transformation)

* TTT = Time - Temperature - Transformation (also IT or C-curve or S-curve) diagram: shows time needed for austenite to transform isothermally into other phases at a constant temperature below A1. Obtained from many small samples quenched into salt baths at different temperatures, held for various times, and quenched; microstructure / hardness / dilatometry used to find fraction transformed.



* LEFT curve = start of transformation (about 1% transformed), RIGHT curve = finish (99%). Left of start curve: only unstable austenite. Between: austenite + product. Right of finish: all product.
* Eutectoid steel: 727 - 550 C: PEARLITE (coarse at high T, fine at lower T; hardness 15 - 40 HRC). Nose at about 550 C (minimum incubation time about 1 s). 550 - 250 C: BAINITE (upper bainite - feathery, lower bainite - needle like; 40 - 55 HRC; tough). Below Ms (~ 220 C for eutectoid steel): MARTENSITE (athermal; depends only on temperature, not time).
* Cooling curves superposed on TTT diagram: slow cooling (annealing) -> coarse pearlite; moderate (normalising) -> fine pearlite; faster that misses the nose (> CCR) -> martensite; interrupted quenching -> bainite.
* Hypo / hypereutectoid steels have an extra curve (proeutectoid ferrite / cementite) above the pearlite nose. TTT is valid ONLY for isothermal

hold; for continuous cooling use CCT diagram (curves shifted to right and down).

Variations in hardening using TTT

* MARTEMPERING (marquenching): quench austenite to just above Ms in hot oil / salt, hold till temperature equalises, then air cool through the Ms-Mf range -> martensite with less distortion, cracking. Then temper.
* AUSTEMPERING: quench to 250 - 400 C (above Ms), hold isothermally until all austenite becomes BAINITE, cool in air. Gives good toughness and ductility with high hardness, no quench cracks, no tempering needed (springs, small tools).

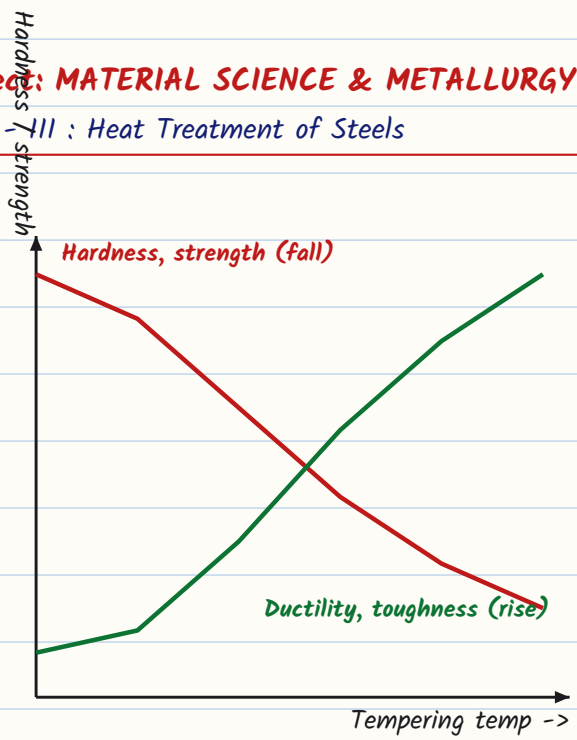
7. Tempering

* Reheating hardened (martensitic) steel to a temperature BELOW A1 (150 - 650 C), holding, and cooling (usually air). It reduces brittleness and internal stresses and improves toughness and ductility at the cost of some hardness / strength.

Temp (C)	Stage / structure	Properties and use
150 - 250	1st stage: epsilon carbide, tempered martensite	Hard (60 HRC), slight toughness - cutting tools, files, razors
250 - 350	2nd stage: retained austenite decomposes to bainite	Spring steel, chisels (400-450 for springs)
350 - 450	3rd stage: cementite precipitates, troostite	Hardness 40-50 HRC, good toughness - springs, hammers
450 - 650	Cementite coarsens, sorbite; spheroidised	Max toughness, 25-35 HRC - gears, shafts, bolts, axles (QUENCHED + TEMPERED = 'hardened and tempered', 'bonnet')

* TEMPER EMBRITTLEMENT: loss of toughness on tempering at 250 - 400 C ('blue brittleness' / 300 C embrittlement) and on slow cooling through 450 - 600 C (impurities P, Sb, Sn segregate to grain boundaries; avoided by adding Mo and by quick cooling).
* SECONDARY HARDENING: Cr, Mo, V, W steels harden again at 500-600 C due to alloy carbide precipitation (red hardness of HSS).

Hardening + Tempering = Quenching and tempering (Q & T) - the most important heat treatment for strong, tough steel parts.



Microconstituents of steel – quick table

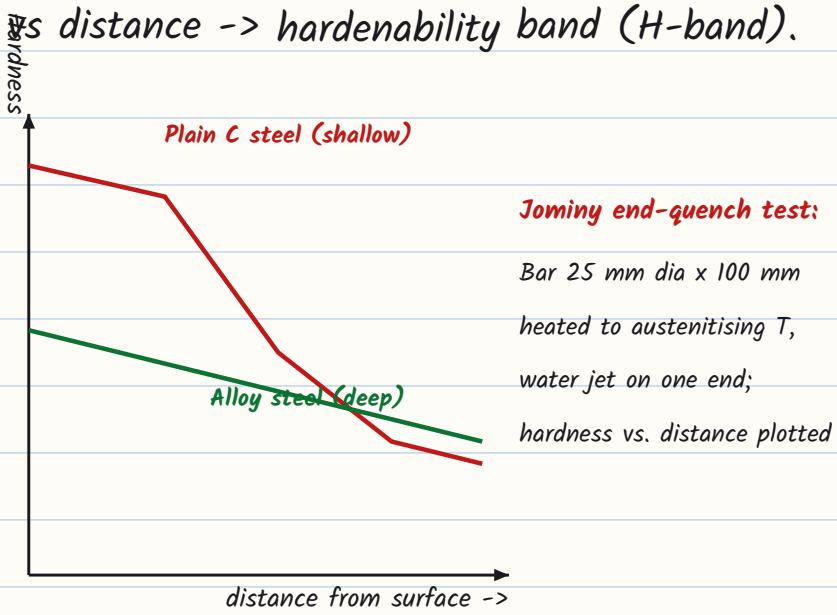
Constituent	Formation	Structure	Hardness (approx.)
Ferrite	Slow cooling	BCC, soft	80-100 HB
Pearlite (coarse / fine)	Eutectoid transformation 727-550 C	Ferrite + Fe3C lamellae	200-300 HB
Spheroidite	Long hold below A1	Cementite spheroids	150-200 HB (soft)
Bainite	Isothermal 550-250 C	Ferrite + fine carbide	40-55 HRC
Martensite	Quench below Ms	BCT supersaturated	55-67 HRC
Troostite (tempered 400)	Tempering	Fine carbide in ferrite	40-50 HRC
Sorbite (tempered 600)	Tempering	Coarser carbide	25-35 HRC

* Retained austenite (Mf below room temp. for > 0.5% C steels) is soft, unstable, causes dimensional change with time (aging) - removed by

sub-zero treatment or double tempering.

8. Hardenability

* Hardenability = ability of steel to be hardened to a depth by quenching (depth of martensite formation); NOT the maximum hardness (hardness depends on %C). It is governed by the position of the TTT nose - alloys shift it right.
* JOMINY END-QUENCH TEST (ASTM A255): standard bar 25 mm dia. x 100 mm long, austenitised, placed in fixture and sprayed with water jet at one end. After quenching, 2 flats are ground and HRC hardness measured at every 1.6 mm distance from quenched end. Plot hardness vs distance -> hardenability band (H-band).



* Distance at which hardness falls to 50% martensite hardness = 'ideal critical diameter' D_I. Alloy steels (Cr, Mo, Ni, Mn) show flat curves = high hardenability; plain carbon steel falls steeply.
* Factors: carbon content, alloy elements (except Co), austenitic grain size (coarse = higher), homogeneity of austenite, quenching medium (severity H value), section size (mass effect).

Quenching media and severity

Medium	Severity H	Used for	Remarks
Brine (10% NaCl)	2.0 - 5.0	Plain carbon steel, simple shapes	Very fast, high risk of cracks
Water	1.0	Low alloy and carbon steels	Fast, distortion risk
Oil (mineral)	0.25-0.7	Alloy steels, tools	Slower, less distortion, fire risk
Polymer solution	0.3-1.0	Induction / large parts	Adjustable severity
Air / gas	0.02	High speed steels, high alloy	Very slow, minimum distortion
Molten salt / lead	Variable	Austempering, martempering	Uniform, no vapour film

* Hot working (hardening) furnaces: box, pit, salt bath, vacuum, fluidised bed, induction. Protective atmospheres: endothermic gas (CO, H2, N2), nitrogen based, to prevent scaling and DECARBURISATION.
* Typical heat treatment of a gear: normalise forging (to refine grain) -> machine -> carburise -> quench in oil -> sub-zero -> temper at 180 C -> finish grind.

9. Surface Hardening Methods

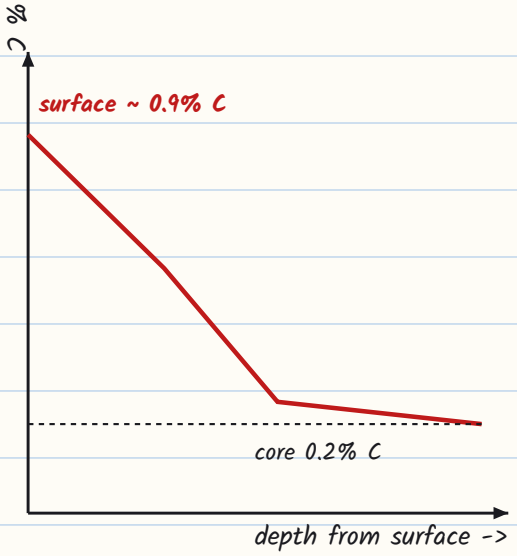
Aim: HARD wear resistant case (surface) with TOUGH shock absorbing core. Two ways: (A) change surface composition (case hardening), (B) change only structure by selective heating and quenching.

A. Carburising (case hardening) - low carbon steel (0.1-0.25% C)

* Pack carburising: parts packed in iron box with charcoal + 10-20% BaCO3 energiser, 900 - 950 C for 6 - 12 h; case depth 0.5 - 1.5 mm.

Gas carburising: furnace atmosphere of CH4 / endo-gas (CO + H2), better control; Liquid carburising: cyanide salt bath (NaCN) 850-900 C, case depth 0.1-0.5 mm (shorter time).

* After carburising: parts are quenched (direct or after re-heating) and tempered at 150 - 200 C. Surface: high carbon martensite (0.8-1% C, 60+ HRC); core: low carbon ferrite-pearlite tough. Uses: gears, camshafts, piston pins, chain links.



B. Nitriding - alloy steel containing Al, Cr, Mo, V (Nitralloy)

* Steel (already hardened and tempered) heated at 500 - 550 C in ammonia (NH3) gas for 20 - 100 h; nascent nitrogen forms very hard nitrides (AlN, CrN, Fe4N). NO quenching, so little distortion.
* Case depth 0.2 - 0.7 mm, hardness 900 - 1100 HV (above carburising), good fatigue, wear and corrosion resistance, keeps hardness up to 500 C. Uses: crankshafts, cylinder liners, spindles, gears, dies. Ion (plasma) nitriding is a modern process.

C. Cyaniding and carbonitriding

* CYANIDING: liquid bath of NaCN + NaCl + Na2CO3 at 800 - 870 C; steel absorbs both C and N; case 0.1 - 0.3 mm in 30 min; quench in oil / water. Used for small parts - screws, nuts, bolts. Toxic bath.

* CARBONITRIDING: gas process with NH3 + hydrocarbon at 750 - 850 C; safe alternative to cyaniding.

D. Flame hardening

* Surface of medium carbon steel (0.4 - 0.6% C) heated by oxy-acetylene flame above A3 and immediately quenched by water spray; case 2-6 mm; no change in surface composition. Good for large parts, gear teeth, lathe beds, rails; parts then tempered.

E. Induction hardening

* Part placed in copper coil carrying high frequency current (10 - 500 kHz) -> eddy current heats the skin rapidly (seconds); quench by water spray. Depth controlled by frequency (higher f -> thinner case). Fast, clean, automatic, repeatable. Uses: crankshaft journals, gears, axles, camshafts, valve seats.

Process	Steel	Temp (C)	Case depth	Hardness
Carburising	0.1-0.25% C	900-950	0.5-1.5 mm	58-62 HRC
Nitriding	Al, Cr, Mo alloy	500-550	0.2-0.7 mm	900-1100 HV
Cyaniding	Low C	800-870	0.1-0.3 mm	55-60 HRC
Flame	0.4-0.6% C	> A3	2-6 mm	50-60 HRC
Induction	0.4-0.6% C	> A3	1-5 mm	50-60 HRC

10. Age (Precipitation) Hardening

* Used for NON-ferrous alloys (Al-Cu duralumin, Al-Mg-Zn, Cu-Be, Ni super alloys, Mg alloys) whose solid solubility DECREASES with falling temperature (solvus line). Discovered by Wilm (1906) in duralumin.
* STEP 1 - SOLUTION TREATMENT: heat to single phase alpha region (e.g. Al-4% Cu at 500 C), soak to dissolve all CuAl2 in alpha.

* STEP 2 - QUENCHING in water: gives supersaturated solid solution (SSSS), soft and unstable.
* STEP 3 - AGEING: natural (room temperature, 4-5 days for duralumin) or artificial (100 - 200 C for several hours). Fine coherent precipitates (GP zones -> theta" -> theta' -> equilibrium theta CuAl2) form; strain fields block dislocations -> hardness and strength increase.
* OVERAGEING: long time / high temperature -> precipitates coarsen, coherency lost, hardness drops. Peak hardness at optimum time.

SSSS --> GP zones --> theta" --> theta' --> theta (CuAl2)

* Mechanism: precipitate particles obstruct dislocation motion (cutting or Orowan bowing). Maraging steels and PH stainless steels also age harden.

II. Cryogenic Treatment

* Deep cooling of material to very low temperature (-80 C shallow, -196 C in liquid nitrogen) slowly, holding 24 hours, and slow warming, usually followed by tempering. An extension of the hardening process.
* Effects: converts RETAINED AUSTENITE to martensite, precipitates fine eta-carbides (increase wear resistance), relieves residual stress, improves dimensional stability. Increases wear resistance 100 - 600%, hardness 1-3 HRC.
* Uses: cutting tools (HSS drills, milling cutters), dies, gauges, bearings, brake discs, gun barrels, engine parts, musical instruments. Never use rapid cooling (cracks).

Important questions

* 1. Effect of alloying elements on Fe-Fe3C diagram (austenite / ferrite stabilisers, eutectoid shift).
* 2. Annealing vs normalising: types, temperature range, structure and applications.
* 3. Draw a TTT diagram for eutectoid steel; mark Ms, Mf; show cooling curves for annealing, normalising, hardening, austempering, martempering.
* 4. Define hardenability; explain the Jominy end quench test.
* 5. Compare carburising, nitriding, cyaniding, flame and induction hardening.
* 6. Explain age hardening of Al-Cu alloy; what is overageing?

Revision: Annealing - soft; Normalising - fine and uniform; Hardening - martensite; Tempering - toughness; Case hardening - hard skin, tough core; Age hardening - non-ferrous, precipitate; Cryogenic - retained austenite removal.

UNIT - IV : POWDER METALLURGY

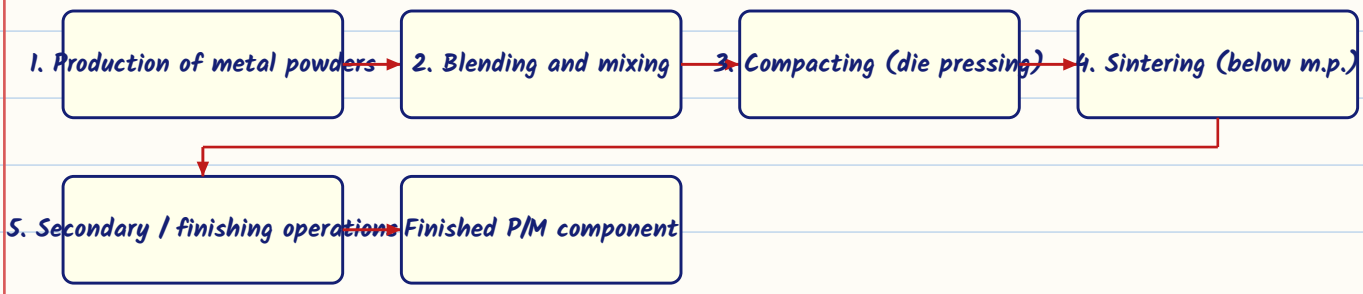
I. Introduction

* POWDER METALLURGY (P/M) = art and science of producing metal powders and making useful engineering components by COMPACTING the powders and SINTERING them (heating below the melting point) - without melting the main constituent.
* History: used by Egyptians (iron tools, 3000 BC), Incas (platinum), 1909 Coolidge made tungsten lamp filaments by P/M; 1930s cemented carbides, porous bearings; today automobile parts form the largest market.

Why powder metallurgy ?

* Metals of VERY HIGH melting point (W 3410 C, Mo, Ta) are difficult to cast; alloys of metals that do not mix in the liquid state (Cu-Pb, W-Cu, Fe-Cu) or metal + non-metal composites (WC-Co, metal-graphite) can be made only by P/M.
* Near net shape parts with close tolerance (+- 0.025 mm) and good surface finish; little or no machining; material utilisation up to 95-97%; controlled porosity (filters, self-lubricating bearings); mass production of small complex parts.

Basic steps of P/M process

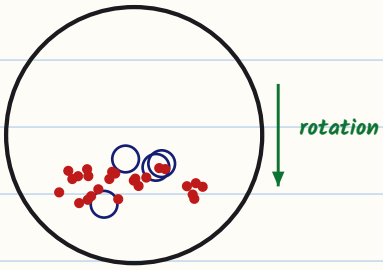


2. Methods of Producing Metal Powders

Properties of powder depend on the method: particle size, shape, purity, apparent density, flow rate, compressibility.

(a) Milling (mechanical pulverisation / crushing)

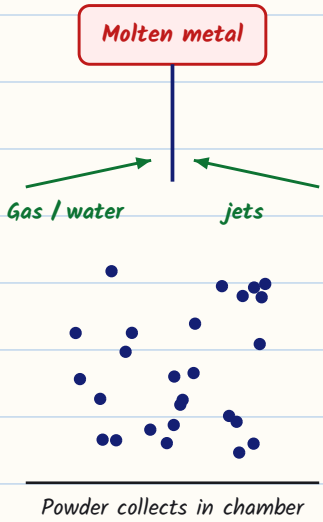
* Brittle metals (Sb, Bi, Cr, Mn, ferro-alloys, oxides, hard metals) and also hard brittle scrap are crushed and ground in BALL MILLS, ROD MILLS, hammer mills, vibratory mills, attritors. Ductile metals get flaky powder (Eddy mill, stamp mill, Hametag process).
* BALL MILL: rotating cylindrical drum partly filled with steel / tungsten carbide balls (30-50% of volume); critical speed N_c = 42.3/sqrt(D) rpm; working speed about 65-75% of N_c. Impact + attrition of balls break the charge. Wet or dry, with inert atmosphere (Ar) for reactive metals. Mechanical alloying (high energy milling) produces ODS alloys.
* Powder: angular, irregular; contamination from the balls; work hardened; cheap but slow process.



Ball mill

Low speed: balls roll / cascade (attrition, fine powder).
Correct speed: balls fall in a cataract - impact breaks charge.
Above Nc balls stick to the wall (centrifuging) - no milling.

(b) Atomization (most widely used - for Al, Fe, steels, Cu, Sn, Zn, P



Atomization

Melt stream broken by high pressure gas / water jets into fine droplets that freeze as powder.
Particle size 10 - 300 micron;
water -> irregular, gas -> spherical.

* Liquid metal stream (from tundish through a nozzle) is disintegrated into droplets by a high pressure jet of water, gas (N2, Ar, air) or by centrifugal force. Droplets solidify quickly into powders.
* Water atomization: irregular shape, cheap, higher oxygen. Gas atomization: spherical, clean, high flowability, costly (super alloys, Ti). Centrifugal / rotating electrode (PREP) : very clean spherical powder. Ultrasonic / vacuum atomization also used. Rapid solidification gives fine microstructure.

(c) Granulation

* Molten metal poured through a screen into water (or stirred while

cooling) - gives coarse granules or shot, used for low melting metals (Zn, Sn, Pb, Cu, Al, Cd). Particles are rough and irregular; used for making pellets, shot and raw material for secondary processes.

(d) Reduction of oxides

* Metal oxide (mill scale, iron ore, tungsten oxide) is reduced in solid state at temperature below m.p. using a reducing gas (H2, CO, cracked ammonia) or carbon: MeO + H2 -> Me + H2O.
* Used for Fe, Cu, W, Mo, Ni, Co. $Fe_2O_3 + 3\ CO \rightarrow 2\ Fe + 3\ CO_2$ (Hoganas sponge iron process). Powder is porous, SPONGY (sponge iron) with good compressibility and green strength. Particle size is controlled by oxide size and temperature. Cheap, high production.

(e) Electrolytic deposition

* Metal ions from an electrolyte are deposited at the cathode at high current density as spongy / powdery / brittle deposit, which is scraped, washed, dried, annealed and milled. Cu powder (CuSO4 + H2SO4 bath), Fe, Ni, Zn, Cr, Mn, Ag powders.
* Very pure (99.5-99.9%), dendritic powders with excellent green strength; but high cost and energy.

(f) Other methods

* Chemical: thermal decomposition of carbonyls, e.g. Fe(CO)5 -> Fe + 5 CO (very pure, spherical, micron-sized carbonyl iron and nickel powders); precipitation from solutions (hydro-metallurgy); hydride-dehydride (Ti, Zr); plasma rotating electrode process; Spray deposition.

Method	Powder shape	Metals	Remarks
Milling	Angular / flaky	Brittle metals, oxides, carbides	Cheap, contamination
Atomization	Spherical (gas), irregular (water)	Al, Fe, Cu, Sn, steels, alloys	Most common; alloy powders
Granulation	Coarse irregular	Zn, Sn, Pb, Cu	Low m.p. metals
Reduction	Spongy, porous	Fe, W, Mo, Cu, Ni	Cheap, mass production
Electrolysis	Dendritic, pure	Cu, Fe, Ni, Zn	High purity, costly
Carbonyl	Spherical fine	Fe, Ni	Very pure

3. Characterisation of Metal Powders

* PARTICLE SIZE & DISTRIBUTION: sieve analysis (> 45 micron, mesh no.), sedimentation, microscopy, laser diffraction (very fine).
  PARTICLE SHAPE: spherical, irregular, dendritic, flaky, angular - affects flow and packing.
* APPARENT DENSITY (loose powder mass / volume, Hall flowmeter funnel), TAP DENSITY, FLOW RATE (time for 50 g to pass through a standard funnel, s/50 g), compressibility (green density vs pressure), green strength, sintering activity, chemical purity (hydrogen loss test for oxide content).

Useful relations

$$\text{Porosity } \% = (1 - \rho_{sintered} / \rho_{theoretical}) \times 100$$

Apparent density = mass of loose powder / volume; Flow rate = time (s)

Theoretical density of mixture: $1/\rho = w_1/\rho_1 + w_2/\rho_2$

Eg: Cu-10% Sn bronze (ρ_{Cu} = 8.96, ρ_{Sn} = 7.3 g/cc): 1/rho = 0.9/8.96 + 0.1/7.3 = 0.1141 -> rho = 8.76 g/cc. If sintered density is 7.5 g/cc, porosity = (1 - 7.5/8.76) x 100 = 14.4%.

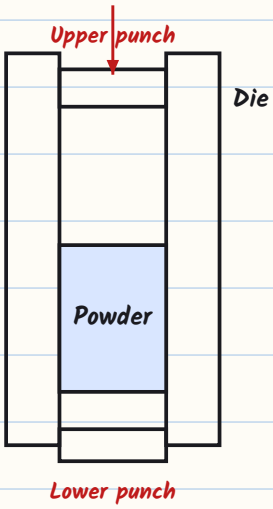
4. Blending and Mixing

* Powders of different metals / sizes are mixed uniformly in V-blender, double cone, drum or Y-mixer for 5 - 30 min. LUBRICANT (0.5 - 1.5% zinc stearate, stearic acid, wax) added to reduce die wall friction and ejection force; binders (for fine powders), additives like graphite for steel.
* Blending = mixing of powders of the same chemical composition (different sizes); Mixing = combining powders of different materials. Over-mixing may cause segregation.

5. Compacting Methods

Aim: to give the loose powder the shape, dimensions, density and 'green strength' (strength to handle before sintering). The pressed part is called GREEN COMPACT.

(a) Conventional die compaction (cold pressing)



Die compaction

Pressure 100 - 700 MPa
Green compact: 70-90% dense, strong enough to handle.
Density gradient due to die wall friction -> double ended pressing / lubricant (zinc stearate).

* Die filled with measured powder (fill depth > compact height, ratio 2:1 to 3:1) -> pressed by upper and lower punches in mechanical, hydraulic or rotary presses -> compact ejected. 100 - 700 MPa, 10 - 300 parts / min.
* Green density about 80-90% of theoretical; pressure raises density; friction gives density variation (corners low). Spring back occurs after ejection (0.1 - 0.3%).

Stages of compaction and effect of pressure

* Stage 1: particles rearrange and fill voids (low pressure). Stage 2: particles deform elastically and plastically, cold weld at contacts, density rises fast. Stage 3: bulk compression, work hardening, density saturates near 90-95% (pores still remain).
* Factors: particle size / shape (irregular gives higher green strength), hardness of powder (soft Cu compacts well, hard W poorly), lubricant, pressing speed, pressure, die design, powder friction (density gradient: use double action pressing or floating die), L/D ratio of part (< 3 recommended).
* Design rules for P/M parts: simple shape that can be ejected from the die; uniform section thickness, avoid thin walls (< 1.5 mm) and sharp

corners, no undercuts or threads perpendicular to pressing direction, generous radii, tapered, minimum hole diameter 2 mm.

(b) Isostatic pressing

* COLD ISOSTATIC PRESSING (CIP): powder in flexible rubber mould in fluid, pressure applied from all directions (200 - 400 MPa) - uniform density, long slender shapes, tubes, large parts.
* HOT ISOSTATIC PRESSING (HIP): pressure 100 - 200 MPa with argon gas at 900 - 1250 C; compaction and sintering together -> near 100% density; super alloy and Ti parts, tool steels, eliminating casting porosity.

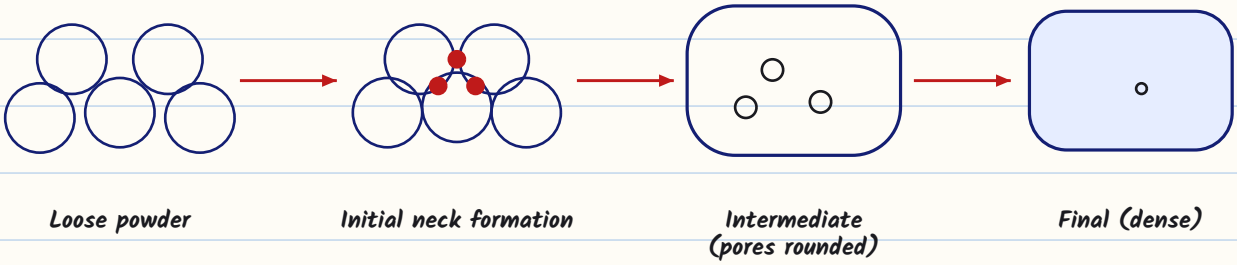
(c) Other compaction methods

* POWDER ROLLING: powder fed between rolls -> continuous strip (Cu, Ni, steel sheet); EXTRUSION: powder in can or with binder extruded; INJECTION MOLDING (MIM): fine powder + 40% polymer binder injected in mould, debinded then sintered - complex small parts (watch cases, medical tools).
* SLIP CASTING (ceramics / metals): slurry poured into plaster mould; HOT PRESSING: pressing and sintering simultaneously in graphite dies (WC, diamond tools); Spark plasma sintering (SPS); forging of sintered preforms (powder forging) ; ADDITIVE MANUFACTURING (laser sintering).

6. Sintering

* SINTERING = heating the green compact in a CONTROLLED ATMOSPHERE at a temperature BELOW the melting point of the main constituent (typically 0.7 - 0.9 T_m; Fe 1120 C, Cu 850 C, Bronze 800 C, WC-Co 1450 C, W 2200 C) for 20-60 min to bond the particles by

atomic diffusion.



Stages of sintering

* 1. Initial stage: particle contact points weld, NECKS form by surface diffusion / vapour transport. 2. Intermediate: necks grow, pores become smooth interconnected tubes, shrinkage and densification (grain boundary and volume diffusion). 3. Final: pores close and become spherical and isolated, slow densification, grain growth.
* Driving force: reduction of total surface energy (surface area) of fine particles. Mechanisms: evaporation-condensation, surface diffusion, volume diffusion, grain boundary diffusion, plastic flow.

Types

* SOLID PHASE sintering: single powder, or mixed powders all remain solid (Fe, Cu, bronze).
* LIQUID PHASE sintering: a lower melting component melts and wets the solid (e.g. WC + Co at 1450 C; Fe + Cu infiltrated; Cu-Sn bronze; W-Ni-Fe) -> faster densification, high density, strong.

Furnace and atmosphere

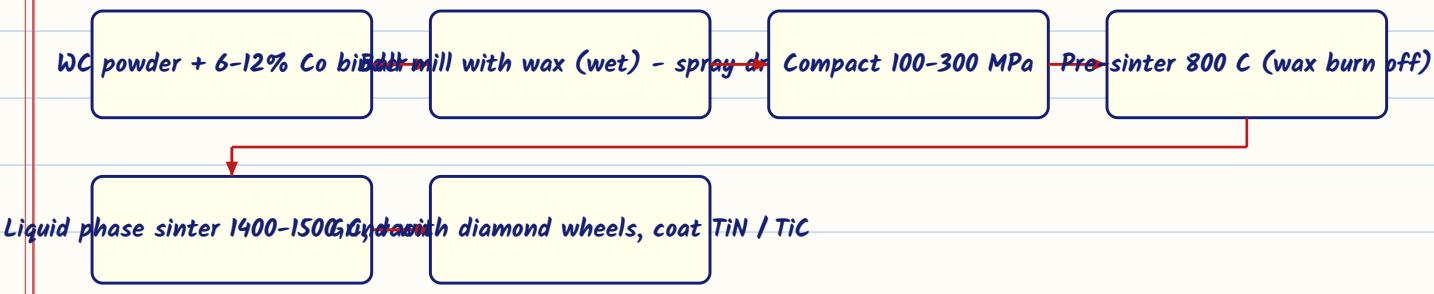
* Continuous mesh belt, walking beam, pusher or batch / vacuum furnaces with 3 zones: pre-heat / burn-off (delubrication at 300-600 C), high heat (sintering), and cooling zones (water jacket, protective atmosphere maintained).

* Atmospheres: endothermic gas (CO, H2, N2), dissociated ammonia (75% H2 + 25% N2), hydrogen, nitrogen-hydrogen, vacuum - to prevent oxidation, reduce surface oxides, control carbon. Never air.
* Variables: temperature, time, atmosphere, heating / cooling rate, green density, particle size. Effects: strength increases (about 2-3 times), density rises, dimensional change (shrinkage or growth 0.1-2%), electrical and thermal conductivity improve, ductility and toughness improve.

7. Methods of Manufacturing Sintered Parts

Route	Steps	Typical parts
Press and sinter (single)	Mix -> press -> sinter	Gears, bushes, cams, structural parts (most common, 70-90% density)
Press - sinter - re-press (re-sizing / coining)	Second pressing improves dimension, density	Close tolerance parts, bearings
Double press - double sinter	Press, pre-sinter, repress, final sinter	High density 92-95%
Infiltration	Sintered porous skeleton filled with molten lower melting metal (Cu into Fe, Ag into W)	Electrical contacts, high strength gears
Impregnation	Pores filled with oil (30 min. vacuum), polymer, wax	Self lubricating bearings
HIP / hot pressing / powder forging	Pressure + temperature	Full density, critical aerospace and tool parts
MIM	Fine powder + binder -> inject -> debind -> sinter	Small complex parts, medical, watch, firearms

Case study 1 : Manufacture of cemented tungsten carbide tools



* Hardness 1400-1800 HV, retains hardness up to 1000 C, used for cutting tool inserts, drills, mining bits, wire drawing dies. Cobalt wets WC during sintering (liquid phase), giving strong, tough compact. Cermet class of material.

Case study 2 : Self lubricating bearings

* Cu (90%) + Sn (10%) bronze powder + graphite 1%, press to 6.4-6.8 g/cc, sinter at 800 C, size, then vacuum impregnate with oil (about 20% by volume). Oil is drawn to the surface by heat of friction and returns by capillary action. Used in fans, domestic appliances, automotive starters, wipers.

Case study 3 : Friction materials, magnets, contacts

* Brake and clutch linings: Cu / Fe matrix + friction modifiers (silica, graphite). Alnico, ferrite and Nd-Fe-B magnets pressed in a magnetic field and sintered. Electrical contacts: Ag-W, Cu-W, Ag-CdO: pressed, sintered and infiltrated (good arc erosion, high conductivity).

8. Secondary Operations

* SIZING / COINING (re-pressing): improve dimensional accuracy and density; MACHINING (drilling, turning, tapping) - limited by pores;

carbide tools used.

* IMPREGNATION with oil (self lubricating bearings hold 10-30% oil), resin (leak sealing, pressure tightness) ; INFILTRATION with Cu; HEAT TREATMENT: steam treatment (blue oxide film Fe3O4 for wear and corrosion), carburising / hardening, case hardening, tempering.
* PLATING and coating (Cu, Ni, Zn, Cr), deburring / tumbling, grinding, welding / joining (sinter bonding), machining, polishing, forging.

9. Applications of P/M Products

* Self lubricating (oil-impregnated) bronze and iron bearings (Oilite) - bushes for home appliances, motors, auto; porous filters (bronze, stainless, Ni, Ti) for fuel, oil, chemical and air.
* Cemented carbide cutting tools (WC + 6-12% Co), dies, mining and drilling bits, wire drawing dies; diamond impregnated tools; ceramic cutting tool inserts.
* Automobile: gears, connecting rods, cams, valve guides, valve seats, synchro hubs, oil pump gears, ABS sensor rings, clutch plates, brake pads (friction materials), piston rings; electrical contacts (W-Cu, Ag-W, Ag-CdO), brushes (Cu-graphite) and relays.
* Refractory metals: tungsten filaments, Mo, Ta capacitor anodes, rocket nozzle parts; Magnets: Alnico, ferrites, rare-earth (Nd-Fe-B, Sm-Co) magnets; heavy alloys (W-Ni-Fe) for counterweights and radiation shields; nuclear fuel, aerospace super alloy discs; surgical implants (porous Ti); powder metal gears for tools; filters.

Advantages and limitations

Advantages	Limitations
Near net shape, minimum waste (97% material use)	Costly metal powders and tooling (dies)
Close tolerances, good finish, no machining	Size limited (press capacity, usually < 2.5 kg)
Controlled porosity, unique combinations	Lower strength / toughness than wrought due to porosity
Suited to very high melting point metals	Complex shapes (undercuts, re-entrant) difficult
Mass production, low labour, no pollution	Economic only for large production volume (> 10,000)

Comparison with casting

Powder metallurgy	Casting
Very high melting metals possible	Limited by furnace / mould capability
Solid state process; good uniformity	Liquid state; segregation, shrinkage, gas defects
Porosity 5-25% (controlled)	Usually dense (pores are defects)
High tooling cost - for mass production	Moulds cheap - good for single / big parts
Accuracy +- 0.025 mm	Accuracy +- 0.25 mm and more

Important questions

* 1. Define powder metallurgy; explain the steps with a block diagram; advantages and limitations.
* 2. Explain atomisation, reduction of oxides and electrolytic deposition methods of powder production.
* 3. Describe milling in a ball mill; critical speed of the mill.

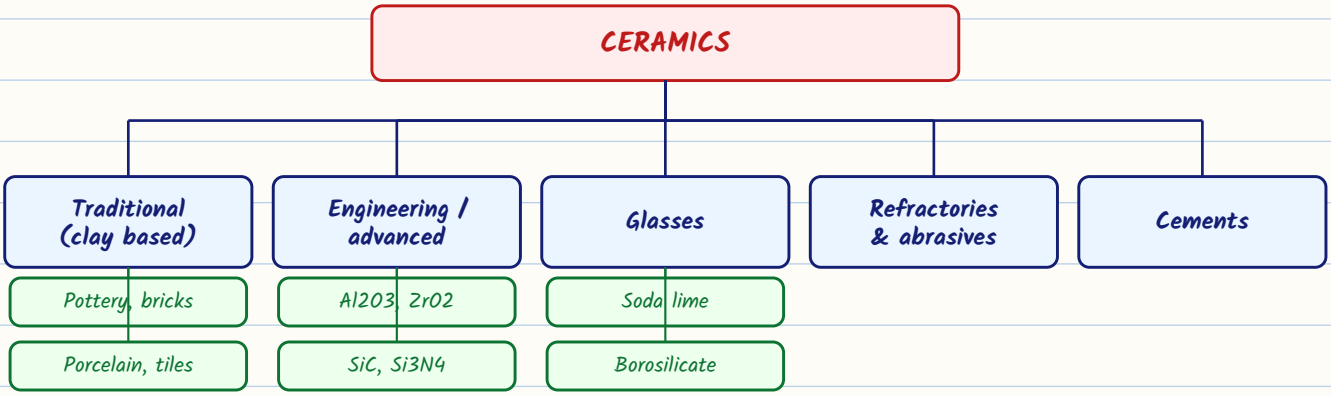
* 4. What is sintering? Stages, atmosphere, variables, types of sintering.
* 5. Write short notes on isostatic pressing, infiltration, impregnation, MIM.
* 6. List applications of P/M parts with suitable examples.

Revision: P/M steps - powder production, blending, compaction, sintering, finishing. Powder methods - milling, atomization, granulation, reduction, electrolysis. Sintering temp = 0.7 - 0.9 Tm in protective atmosphere.

UNIT - V : CERAMIC AND ADVANCED MATERIALS

1. Ceramics - introduction

* CERAMICS = inorganic, non-metallic solids (compounds of metals with O, C, N, B, Si - oxides, carbides, nitrides, borides) processed at high temperature. Bonding is IONIC and / or COVALENT.
* General properties: high hardness and compressive strength, very high melting point, chemical inertness / corrosion resistance, wear resistance, electrical and thermal INSULATION (some are semi / super conductors), low density, but BRITTLE (low tensile strength and toughness) and poor thermal shock resistance.
* Why brittle? Ionic / covalent bonds are directional, very few slip systems; like-charged ions cannot slide past each other; pores and micro-cracks act as stress raisers (Griffith flaw: $\sigma_f = (2 E \gamma / \pi a)^{1/2}$).



2. Crystalline Ceramics

* Crystal structure follows ionic radius ratio r_c/r_a (cation / anion) and charge neutrality. Cations are generally smaller than anions and fit in the interstices of closely packed anions.

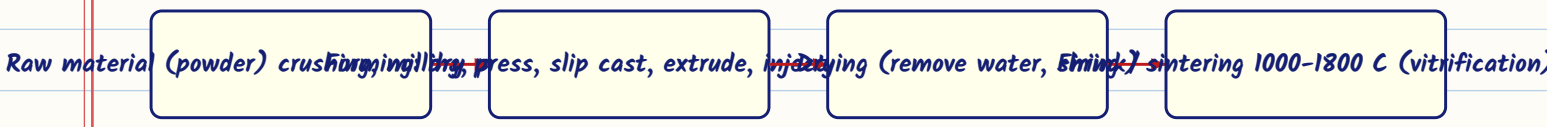
r(c)/r(a)	Coordination number	Structure examples
< 0.155	2 (linear)	-
0.155 - 0.225	3 (triangular)	B2O3
0.225 - 0.414	4 (tetrahedral)	ZnS (zinc blende), SiO2
0.414 - 0.732	6 (octahedral)	NaCl (rock salt), MgO, FeO
0.732 - 1.0	8 (cubic)	CsCl, CaF2 (fluorite) ZrO2

* AX type: rock salt (NaCl, MgO, NiO), CsCl, zinc blende. AmXp: fluorite CaF2 (UO2, ZrO2). AmBnXp: perovskite BaTiO3 (ABO3; piezo / ferro-electric, capacitors), spinel MgAl2O4 (AB2O4; ferrites - magnetic). Silicates: SiO4 tetrahedra (silica, quartz, clay minerals, mica, feldspar).

Important engineering ceramics

Ceramic	Properties	Uses
Alumina Al2O3	Hard (HV 1800), strong, insulator, m.p. 2050 C, chemically stable	Spark plug insulator, substrates, cutting tools, bio-implants, refractory
Zirconia ZrO2 (PSZ, YSZ)	High toughness (transformation toughening), low k, ionic conductor	Thermal barrier coating, oxygen sensors, knives, dental crowns, fuel cells
Silicon carbide SiC	Very hard, strength at 1500 C, high k, oxidation resistant	Abrasives, kiln furniture, heating elements, brake discs, armour
Silicon nitride Si3N4	High strength, thermal shock resistance, creep resistance	Bearings, turbine rotors, cutting tools, engine parts
Boron carbide B4C, BN, TiC, TiB2	Extremely hard, light	Armour, nozzles, neutron absorbers, cutting tools
Beryllia BeO	High thermal conductivity, electrical insulator	Electronic substrates (toxic dust)
Barium titanate BaTiO3, PZT	Ferro / piezo electric, high permittivity	Capacitors, transducers, sensors

Processing of ceramics



* Fired ceramic: crystalline grains + glassy phase + pores (porosity reduces strength). Hot pressing and HIP get dense, strong ceramics. Sol-gel and CVD used for films / fibres.

3. Glasses

* GLASS = inorganic AMORPHOUS (non-crystalline) solid obtained by cooling a melt so fast that crystallisation is avoided; a rigid super-cooled liquid. No sharp melting point - softens gradually over a range (Glass transition temperature T_g).
* Structure: random network of SiO4 tetrahedra (network former: SiO2, B2O3, P2O5), network modifiers (Na2O, K2O, CaO, BaO - break the Si-O-Si bonds, lower viscosity and m.p.), intermediates (Al2O3, PbO, TiO2).
* Properties: transparent, hard, brittle, good chemical resistance (except HF), electrical insulator, low thermal conductivity, isotropic, easily shaped when hot (viscous). Characteristic temperatures: strain point, annealing point (~ 10^{13} poise), softening, working and melting points.

Type of glass	Composition	Properties and uses
Soda-lime (window, bottle)	70-72 SiO2, 12-16 Na2O, 10-12 CaO	Cheap (90% of all glass); windows, bottles, bulbs; poor thermal shock
Borosilicate (Pyrex)	80 SiO2, 13 B2O3, 4 Na2O, 2 Al2O3	Low expansion, thermal shock and chemical resistant; lab ware, cookware, tube lights
Lead glass (crystal)	50-60 SiO2, 25-30 PbO, K2O	High refractive index, brilliance; optics, tableware, radiation shielding
Fused silica / vitreous silica	99.9% SiO2	Very low expansion, UV transparent; crucibles, optical fibre, telescope mirrors
96% silica (Vycor)	96 SiO2, 4 B2O3	High temperature use up to 900 C
Glass ceramic (Pyroceram)	Li2O-Al2O3-SiO2 + TiO2 nucleant	Controlled crystallisation (90% crystalline), strong, zero expansion; cooktops, missile radomes
Toughened / tempered	Soda-lime heat treated	Surface compressive stress; car side windows, doors
Laminated (safety)	Glass + PVB interlayer	Wind screens - does not shatter

* Manufacturing: melting (1400 - 1600 C) -> forming by blowing, pressing, drawing, float process (glass floated on molten tin bath - flat sheets), fibre drawing -> ANNEALING (remove residual stresses) -> finishing.
* Tempering: heat above T_g (about 650 C) then cool surface quickly with air jets - surface in compression, core in tension - 4-5 times stronger. Chemical strengthening: K+ ion exchange.

4. Cermets

* CERMET = CERamic + METal composite: hard refractory ceramic particles bonded with metallic binder; combine the hardness, high m.p. and oxidation resistance of ceramics with the toughness and thermal shock resistance of

metals. Made by powder metallurgy (liquid phase sintering).

Cermet	Constituents	Uses
Cemented carbide	WC + 6-25% Co (also TiC, TaC + Ni)	Cutting tool inserts, drill bits, dies, wear parts
Oxide cermet	Al2O3 + Cr (or Fe, Ni, Mo)	Turbine blades, high temperature parts, nozzles
Titanium carbide cermet	TiC + Ni / Mo	High speed finishing tools, cutting
Nuclear cermets	UO2 + Stainless steel / Mo	Fuel elements
Cr3C2 + Ni (Cr)	Chromium carbide + nichrome	Corrosion resistant, coating by thermal spray
MgO / Al2O3 + Ni	Oxides + metals	Brake linings, electrical contacts

* Cermet properties: hardness 1200-1800 HV, compressive strength very high, density 6-15 g/cc, retain hardness at 800-1000 C. Brittle compared with metals but tougher than monolithic ceramics.

5. Abrasive Materials

* ABRASIVES are hard, tough, sharp-edged materials used for cutting, grinding, polishing and lapping other materials (grinding wheels, coated abrasives - sand paper, abrasive powders, lapping paste, blasting).
* Requirements: high hardness (greater than workpiece), toughness, friability (self-sharpening: fractures to expose new sharp edges), thermal stability, chemical inertness.

Abrasive	Hardness (Knoop / Mohs)	Notes and use
Diamond (natural / synthetic)	7000-8000 / 10	Hardest; grinding carbide, glass, ceramics, dressing wheels, polishing; reacts with Fe at high T
Cubic boron nitride (CBN, Borazon)	4500 / 9.9	Grinds hardened steel, HSS, super alloys; stable up to 1300 C
Silicon carbide (Carborundum)	2500 / 9.5	Grinds cast iron, non-ferrous, glass, stone, ceramic - very sharp but brittle (green or black)
Aluminium oxide (alumina, Corundum / emery)	2000 / 9	Most used; steel and alloy grinding; tougher; white, brown, pink (Cr), sol-gel alumina
Boron carbide	2800 / 9.3	Lapping, nozzles, dressing tools
Garnet, emery, quartz, flint, pumice	800-1200	Natural; sand paper, polishing, sandblasting

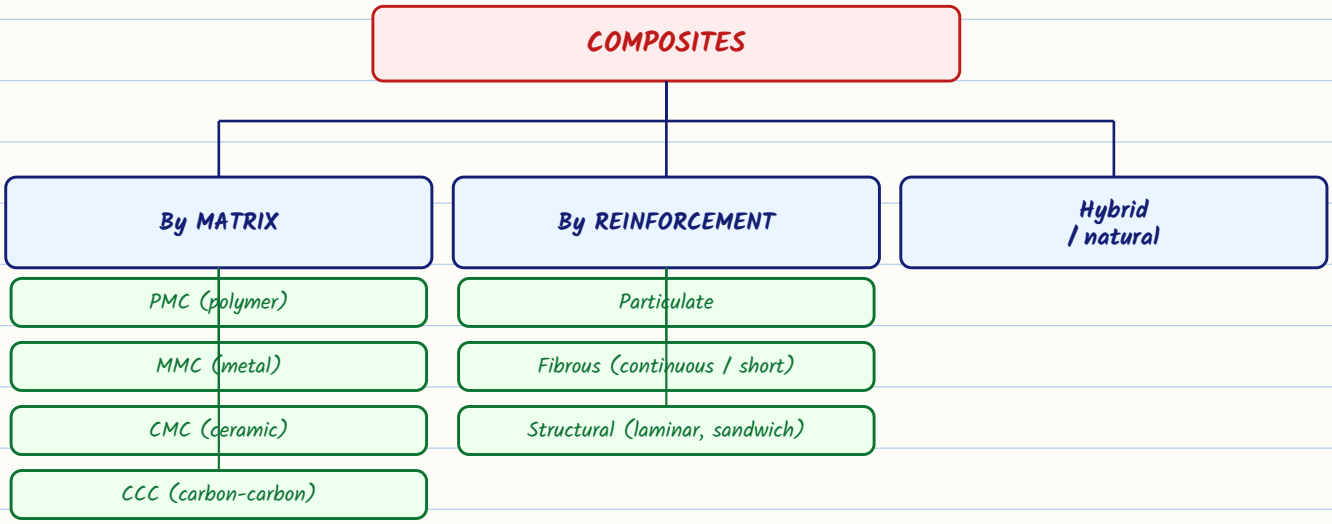
* GRINDING WHEEL: abrasive grains + BOND (vitrified (ceramic clay) - most common, silicate, shellac, rubber, resinoid (bakelite), metal bond (Cu, Ni) for diamond). Wheel marking e.g. A 60 K 5 V: abrasive type, grit size, grade (hardness), structure, bond.

6. Composite Materials

* COMPOSITE = multiphase material of two or more physically distinct and chemically different constituents - the MATRIX (continuous, binds, transfers load, protects) and the REINFORCEMENT (dispersed phase, carries most load) - separated by an interface, giving a combination of properties better than either alone.
* Examples: wood, bone, concrete, fibre glass, plywood, tyre, carbon-epoxy aircraft parts, WC-Co tool.

* Merits: high specific strength and stiffness (strength/density), design flexibility (anisotropy tailored), fatigue and corrosion resistance, light weight, damping, thermal stability. Demerits: costly, anisotropic, difficult to repair, poor out-of-plane strength, delamination, recycling issues.

Classification of composites





Particle reinforced | Fibre reinforced | Structural (laminar)

7. Particle Reinforced Composites

* Reinforcement is hard particles (equiaxed, size 0.01 - 100 micron) in matrix. LARGE particle composites: particle - matrix interaction on macroscale; load shared: concrete (gravel in cement), cermets (WC-Co), carbon black in rubber (tyre), Al-SiC, Ni-Al2O3.

* DISPERSION STRENGTHENED composites: very small particles (10 - 100 nm), 1 - 15 vol% hard oxide (e.g. ThO2 in Ni, Al2O3 in Al (SAP), Y2O3 in super alloys). They block dislocations (Orowan mechanism); matrix carries load; retains strength at high temperature.
* Rule of mixtures for large particles - elastic modulus bounds:

$$\text{Upper: } E_c = E_m V_m + E_p V_p \quad \text{Lower: } 1/E_c = V_m/E_m + V_p/E_p$$

* Uses: tool bits, tyres, grinding wheels, electrical contacts, rocket nozzles, concrete.

8. Fibre Reinforced Composites

* Most important class: strong stiff fibres in a ductile matrix. Fibre is the main load carrier. Mechanical properties depend on (a) fibre-matrix BOND, (b) fibre length (critical length $l_c = \sigma_f d / 2\tau_c$), (c) fibre orientation and (d) fibre volume fraction.
* CONTINUOUS (aligned) fibres: maximum strength along fibre axis (longitudinal), very weak transverse. DISCONTINUOUS (chopped) aligned or random (2-D) - isotropic in plane (sheet moulding compound, SMC). Woven fabrics, mats, 3-D weaves, laminates [0 / 90 / +-45] to give quasi isotropic properties.

Fibres

Fibre	Density g/cc	Tensile strength GPa	Modulus GPa	Remarks
E-glass	2.55	3.4	72	Cheapest, most used (GFRP)
S-glass	2.5	4.5	85	Stronger, aerospace
Carbon (high strength)	1.8	3.5-5.5	230-300	High stiffness, light, brittle (CFRP)
Aramid (Kevlar 49)	1.45	3.6	130	Tough, impact, bullet-proof vests
Boron	2.6	3.5	400	Very stiff, costly (W core)
SiC / alumina	2.5-3.2	2-3.4	200-400	High temperature MMC, CMC
Natural (jute, sisal, flax)	1.3-1.5	0.4-0.8	20-50	Low cost, biodegradable

Rule of mixtures (continuous aligned fibres)

$$E_{cl} = E_f V_f + E_m V_m \text{ (iso-strain, longitudinal)}$$

$$1/E_{c-} = V_f/E_f + V_m/E_m \text{ (iso-stress, transverse)}$$

Eg: 60% E-glass (72 GPa) in epoxy (3 GPa): E_longitudinal = 0.6 x 72 + 0.4 x 3 = 44.4 GPa; transverse: 1/E = 0.6/72 + 0.4/3 = 0.1417 -> E = 7.06 GPa.

9. Polymer, Metal, Ceramic and Carbon-Carbon Composites

(a) PMC - Polymer Matrix Composites

* Matrix: thermosets (epoxy, polyester, phenolic, vinyl ester, polyimide) or thermoplastics (PEEK, nylon, PP, PPS). Reinforcement: glass (GFRP), carbon (CFRP), aramid (AFRP). Service temperature limited (up to 150-300 C).
* Manufacture: HAND LAY-UP / wet lay-up + spray-up (boats, tanks);

VACUUM BAGGING and autoclave moulding of prepregs (aircraft); FILAMENT WINDING (pipes, pressure vessels, rocket motor cases); PULTRUSION (continuous constant section rods, beams, ladders); RESIN TRANSFER MOULDING (RTM, VARTM); compression moulding of SMC / BMC (auto parts); injection moulding (short fibre thermoplastics); prepreg tape laying.

* Uses: aircraft wings and fuselage (Boeing 787 > 50% composite), wind turbine blades, car bodies, boats, sports goods (rackets, golf shafts, bicycles), pipes, printed circuit boards, tanks.

(b) MMC - Metal Matrix Composites

* Matrix: Al, Mg, Ti, Cu, Ni (ductile) reinforced by SiC, Al_2O_3, B_4C particles or whiskers, carbon, boron or SiC fibres. Higher service temperature (to 500 C), higher strength, stiffness, wear resistance, thermal / electrical conductivity than PMC, no moisture absorption; but costlier and heavier.
* Manufacture: STIR CASTING (particles stirred into molten Al, then cast - cheapest), SQUEEZE CASTING / liquid infiltration, POWDER METALLURGY (blend, press, sinter), diffusion bonding (fibre + foil), spray deposition, in-situ reactions, friction stir processing.
* Uses: Al-SiC brake discs and drums, piston and cylinder liners, connecting rods, drive shafts, bicycle frames, electronic heat sinks, aerospace structures (Al-B), turbine blades.

(c) CMC - Ceramic Matrix Composites

* Matrix: Al_2O_3, SiC, Si_3N_4, glass-ceramic, mullite, carbon; reinforcement: SiC fibres / whiskers, carbon fibre, Al_2O_3 fibres, zirconia particles. Fibres are added NOT for strength but to increase TOUGHNESS - crack deflection, fibre pull-out and bridging prevent catastrophic failure.
* Manufacture: hot pressing, CVI (chemical vapour infiltration), reaction bonding, sol-gel, slurry infiltration + sintering, polymer infiltration pyrolysis (PIP), melt infiltration. Uses: jet engine hot sections (shrouds, combustor liners), brake discs (C/SiC in Porsche, supercar), heat shields, cutting tools (Al_2O_3-SiC whisker), armour, gas turbine blades.

(d) CCC - Carbon-Carbon Composites

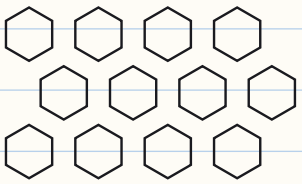
* Carbon fibres in a carbon (graphite) matrix. Made by laying fibres (2-D, 3-D weave) with resin / pitch -> carbonise (800-1000 C) -> repeated DENSIFICATION by impregnation + pyrolysis or CVI -> graphitise (2500 C+).
* Properties: strength and stiffness maintained up to 2000+ C (in inert / protected atmosphere), very low density (1.6-1.9 g/cc), low thermal expansion, high thermal shock resistance, good friction and wear, biocompatible; weakness: OXIDATION above 400-500 C (needs SiC coating).
* Uses: space shuttle nose cone and wing leading edges, rocket nozzles and throats, aircraft and F1 racing car brake discs, hot press dies, furnace fixtures, bone plates and implants.

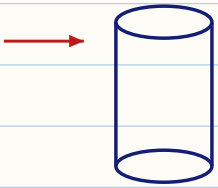
Property	PMC	MMC	CMC	CCC
Max temp	150-300 C	300-600 C	1000-1600 C	2000+ C (inert)
Density	Lowest	Medium	Medium	Low
Toughness	Good	Good	Poor -> improved	Fair
Cost	Low-medium	Medium-high	High	Highest

10. Introduction to Nanomaterials

* NANOMATERIALS: materials with at least ONE dimension in the nanometre range (1 - 100 nm; 1 nm = 10^{-9} m) - properties differ from bulk due to (i) very large SURFACE AREA / volume ratio, (ii) quantum confinement effects.
* Classification by dimension: 0-D (nanoparticles, quantum dots, fullerene C60), 1-D (nanowires, nanotubes, nanorods), 2-D (thin films, graphene, nanoplates), 3-D (nanocrystalline bulk, nanocomposites).

Graphene sheet Rolled -> nanotube





SWCNT dia 0.4-3 nm,
MWCNT 2-100 nm
Tensile strength > 50 GPa
E about 1 TPa, k = 3000 W/mK
Armchair - metallic;
zigzag / chiral - semiconducting

Types and properties

* FULLERENE C60 (buckyball; Kroto, Smalley, Curl 1985): 60 carbon atoms in a soccer ball (20 hexagons, 12 pentagons). CARBON NANOTUBES (Iijima 1991): rolled graphene sheet: SWCNT / MWCNT, very high strength (50-100x steel at 1/6 weight), high conductivity. GRAPHENE: single layer of C atoms (2004, Geim, Novoselov), 2-D, strongest known, very high carrier mobility. Quantum dots (CdSe) - size dependent colour. Nano-Ag, TiO2, ZnO, Au particles.
* Size effects: Hall-Petch strengthening (nanocrystalline metals are 3-5 times harder), melting point falls, ceramics become ductile (superplastic), optical colour changes (Au nanoparticles red), magnetic: superparamagnetism, catalytic activity rises with surface.

Synthesis

* TOP-DOWN: ball milling (mechanical alloying), lithography, etching, severe plastic deformation (ECAP). BOTTOM-UP: chemical vapour deposition (CVD), sol-gel, co-precipitation, laser ablation, arc discharge (CNT, fullerene), electro-deposition, self assembly, inert gas condensation, hydrothermal.

Applications

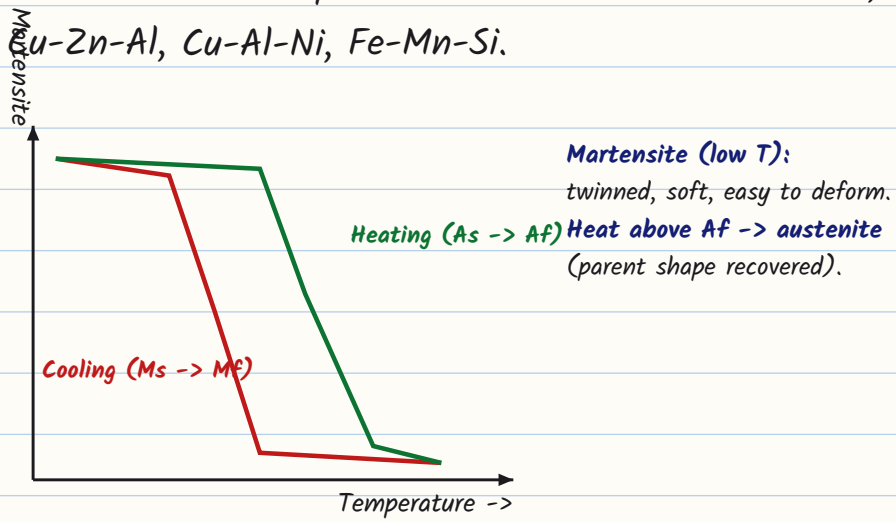
* Electronics (transistors, memory, displays, sensors), medicine (drug delivery, imaging, cancer therapy, antibacterial Ag), energy (solar cells, Li-ion batteries, fuel cells, H2 storage), cosmetics (sunscreen ZnO / TiO2), water purification, catalysts, nano-composites (sports goods), coatings (self cleaning, scratch resistant), textiles. Concerns: toxicity, environmental impact.

II. Smart Materials

* SMART (intelligent / responsive) materials sense a change in environment (stimulus: temperature, stress, electric / magnetic field, light, pH) and respond by a CHANGE of one or more properties (shape, colour, stiffness, viscosity) in a reversible and controllable way. Used as sensors, actuators and adaptive structures.

(a) Shape memory alloys (SMA)

* Alloys that 'remember' original shape: after plastic deformation at low temperature (martensite) they return to the shape on heating above transformation temperature (austenite). NiTi (Nitinol, 55 Ni - 45 Ti), Cu-Zn-Al, Cu-Al-Ni, Fe-Mn-Si.



* Thermoelastic martensite transformation: one-way effect (shape recovered only on heating), two-way effect (trained for both shapes), SUPER-ELASTICITY (pseudo-elasticity: 8% recoverable strain above Af, by stress induced martensite). Uses: orthodontic wires, stents, eye-glass frames, couplings and fasteners, robotic actuators, thermostat valves, aircraft wing morphing, anti-scald valves.

(b) Piezoelectric materials

* Generate voltage when mechanically stressed (direct effect) and deform on applying electric field (converse effect). Quartz, Rochelle salt, BaTiO3, PZT (lead zirconate titanate), PVDF polymer. Uses: ultrasonic transducers, sensors, microphone, lighter, sonar, inkjet, actuators, energy harvesting, accelerometers,

quartz oscillators.

(c) Magnetostrictive and magnetic SMA

* Terfenol-D (Tb-Dy-Fe) changes length in a magnetic field (up to 0.2%); used in sonar, vibration control, precision actuators. Ferromagnetic shape memory alloys (Ni-Mn-Ga) respond to field.

(d) Electro- and magneto- rheological fluids (ER / MR)

* Suspensions of fine particles in oil that change from liquid to near-solid (viscosity up to 10^5 times in ms) when electric or magnetic field is applied. Uses: adaptive shock absorbers and dampers, clutches, brakes, car suspension, prosthetic limbs, seismic dampers.

(e) Others

* PHOTOCHROMIC (glass with AgCl - darkens in UV, spectacles) and ELECTROCHROMIC (smart windows, mirrors) materials; THERMOCHROMIC (colour changes with T - thermometers); Fibre optic sensors; Self healing polymers and concrete; Electro-active polymers (artificial muscles); Hydrogels (pH / temperature responsive); Thermoelectric materials (Bi2Te3: heat <-> electricity); Chemo-chromic.

Smart material	Stimulus	Response	Application
SMA (Nitinol)	Temperature / stress	Shape change	Stents, wires, actuators
Piezoelectric (PZT)	Stress / Electric field	Voltage / strain	Sensors, sonar
Magnetostrictive (Terfenol-D)	Magnetic field	Length change	Actuators
MR / ER fluid	Magnetic / electric field	Viscosity	Dampers, clutches
Photochromic glass	UV light	Colour	Sunglasses

Revision: Ceramics - ionic / covalent, hard, brittle. Glass - amorphous. Cermet - WC + Co. Composites = matrix + reinforcement; PMC, MMC, CMC, CCC. Nano: 1-100 nm, CNT, graphene. Smart: SMA (Nitinol), piezo (PZT), MR fluid.